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Full Length Article

The stress-controlled mechanism of deep-buried tunnels based on functional gradient reinforcement in high tectonic stress zones

Yaocai Ma [a], Hehua Zhu [a,b], Wuqiang Cai [a,*], Chenlong Su [a], Xiangyang Wei [a], Zhi Zheng [c], Jian Pu [a]

[a] Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai, 200092, China
[b] State Key Laboratory for Disaster Reduction in Civil Engineering, Tongji University, Shanghai, 200092, China
[c] Key Laboratory of Disaster Prevention and Structural Safety of Ministry of Education, Guangxi University, Nanning, 530004, China

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ABSTRACT

Grouting reinforcement serves as a critical measure for the stability design and disaster prevention of deep-buried tunnels. However, the grouting reinforcement effect on the unloading stress path of surrounding rock in high tectonic stress zones and its mechanisms for disaster prevention and control remain unclear. This paper proposes a functional gradient bearing concept for grouting reinforcement of fractured surrounding rock in deep-buried tunnels. The functionally graded reinforced zone is designed, exhibiting a radial mechanical property gradient where stiffness increases and strength decreases from shallow to deep regions of the surrounding rock. A semi-analytical method based on the complex variable theory is specifically developed for elastic-plastic mechanical analysis of reinforced surrounding rock. Subsequently, the effects of rock strength and stiffness on the morphological characteristics of the plastic zone around deep-buried circular tunnels in high tectonic stress zones are investigated. The study shows that: (1) The annular grouting reinforcement design has limited effectiveness in preventing the failure of surrounding rock; (2) Grouting reinforcement significantly enhances the bearing capacity of tunnels in weak and fractured rock masses, but excessive grouting to control the plastic zone is not recommended; (3) The bearing capacity of the shallow surrounding rock and the stress regulating function of the deeper surrounding rock can be fully utilized if the functional gradient bearing concept is considered, achieving coordinated bearing between shallow and deep surrounding rock. The research provides new insights for the gradient grouting reinforcement and stress control optimization design of surrounding rock in deep-buried circular tunnels.

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1. Introduction

In general, the distribution of geostress across mainland China exhibits a pattern where tectonic stress dominates in shallow regions, while gravitational stress prevails in deeper regions. However, in the Qinghai-Tibet Plateau, where the Indian and Eurasian plates collide and uplift, intense geotectonic activity has led to a distinct geostress regime, characterized by prominent horizontal tectonic stress (Kang et al., 2010). This has brought unprecedented challenges to deep-earth engineering projects and the construction of the Sichuan-Tibet Railway in China (Cui et al., 2022). A large number of ultra-long, deeply buried tunnels have been constructed along the Sichuan-Tibet Railway. Both physical model tests and numerical simulations clearly demonstrate that the evolution of the surrounding rock stress state governs its fracture mechanisms, resulting in various hazard forms induced by different failure modes around tunnels, particularly under extremely complex stress conditions (Zhang et al., 2024a; Yu et al., 2025). This finding provides important guidance for disaster prevention and mitigation in tunnel engineering. In high tectonic stress zones, excavation unloading inevitably leads to a series of engineering disasters, including macro-scale tensile-shear failures, rock bursts, and large deformations of the surrounding rock (Du et al., 2016; Gong et al., 2019; Chu et al., 2019, 2024; Wang et al., 2023). Traditional

* Corresponding author.
E-mail address: caiwuqiang95@tongji.edu.cn (W. Cai).
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shallow-buried tunnel engineering practices have gradually developed an empirical design system dominated by gravitational stress fields, such as the Terzaghi theory and the Platts theory of ground pressure. However, these methods are difficult to apply to deep-buried tunnels in areas of high tectonic stress, and traditional empirical design methods for shallow engineering face significant challenges (Zhu et al., 2023; Ma et al., 2024). On the one hand, the deformation and failure characteristics of tunnel surrounding rock are stress-dependent (stress-controlled). The disturbed stress distribution of the surrounding rock formed by tunnel excavation varies significantly across different geostress environments. As a result, there are notable differences in the strength characteristics (Si and Gong, 2020; Wu et al., 2023), brittle-ductile transition behavior (Xie et al., 2019), failure modes (Ling et al., 2023; Zhao et al., 2023a), catastrophic tendencies (Meng et al., 2021; He et al., 2021), and dynamic disturbance properties (Feng et al., 2023; Zheng et al., 2024a, 2024b) of surrounding rock in deep-unloading zones under high tectonic stress, compared to shallow-buried tunnels dominated by gravitational stress field. On the other hand, the intense unloading caused by tunnel excavation leads to slippage and opening of pre-existing joints and fractures in the surrounding rock, as well as the formation of secondary discontinuities (Lei et al., 2017). This induces rock fragmentation and failure, further degrading the strength parameters of the surrounding rock (Cai et al., 2022a). Therefore, stability control in deep-buried tunnels in high tectonic stress zones should fully consider the reinforcement of fractured rock in highly disturbed areas, while also employing stress control to proactively prevent and mitigate rock failure.

Grouting reinforcement, a widely adopted method for the stability control of surrounding rock in underground engineering, significantly improves the integrity of rock masses by filling joints and fractures. This reinforcement technique transforms the engineering rock mass into a natural support material, improving its deformation and failure mechanical behavior. Therefore, it is an effective measure for preventing and controlling the stability of tunnel surrounding rock in high tectonic stress zones (Sui et al., 2015). Treating the reinforced zone as supporting material significantly reduces the time-dependent deformation of surrounding rock during the excavation unloading process (Kargar et al., 2020). Currently, there has been extensive research on grouting theory, grouting materials, grouting parameters, and grouting techniques (Gothäll and Stille, 2009, 2010; Kim et al., 2019; Li et al., 2022). However, there is a lack of in-depth understanding of the reinforcement effect on the stress-controlled mechanism of surrounding rock. Grouting reinforcement improves the bearing characteristics of surrounding rock by changing its mechanical properties and fully utilizes the strength potential of the rock mass by optimizing the stress state, enabling the effective transfer and release of stress in the surrounding rock. Quantitatively assessing the mechanical behavior of tunnel surrounding rock and the effectiveness of reinforcement under grouting is crucial for accurate stability analysis and the scientific design of reinforcement parameters. Numerical methods have advantages in investigating the mechanical behavior of the grout-rock interface (Gothäll and Stille, 2009, 2010; Lei et al., 2017), grout diffusion and consolidation patterns (Deng et al., 2018; Xiong et al., 2018), and complex excavation processes (Li et al., 2018; Benato et al., 2023; Xu et al., 2024). However, they are limited to the rapid analysis of surrounding rock stress during grouting reinforcement and in sensitivity studies of reinforcement parameters. Analytical methods can effectively address these shortcomings. In theory, the reinforcement effect of grouting can be represented as an enhancement of the macro-mechanical parameters of the rock mass (Yao et al., 2023; Ma et al., 2023a). Therefore, a simplified mechanical model of the tunnel can be established for the natural rock zone and the reinforced zone based on the concept of an equivalent reinforced zone, thereby facilitating the calculation of the mechanical response of the surrounding rock (Fang et al., 2013; Li et al., 2018; Zhao et al., 2023b). The reinforced zone functions as a distinct material medium, exhibiting mechanical properties that differ from those of the natural rock zone.

Grouting reinforcement enhances the strength and stiffness parameters of the surrounding rock and optimizes the stress control path for its stability. The convergent-constrained mechanical properties or the ground response curves (GRCs) are changed. This enhancement allows for the optimization of the surrounding rock-lining interaction and the adjustment of stress paths, ultimately improving the load-bearing characteristics of the surrounding rock. Fang et al. (2013) conducted a theoretical analysis of the mechanical response of surrounding rock under full-section reinforcement methods. The characteristic of GRCs and the evolution of plastic zone under varying reinforcement parameters and geostress conditions were revealed, indicating that the surrounding rock deformation can be effectively constrained through the reinforcement of the plastic zone. Li et al. (2018) further analyzed the calculation methods for the thickness of the reinforced zone and the recommended values of reinforcement mechanical parameters under the influence of seepage and grouting. Zhao et al. (2023b) considered the post-peak strength drop of the surrounding rock and demonstrated the positive impact of maintaining the peak strength of the surrounding rock after reinforcement on its stability. Sun et al. (2024) found that when using grouted rock bolts for reinforcement, the length of the bolts should ideally exceed the radius of the burst damage zone and the plastic zone to maximize the effectiveness of the support provided by the bolts. This is particularly important when designing grouted rock bolts for tunnels in weak and fractured rock mass, where the burst damage zone effect must be fully considered. Similarly, this method can also be applied to the mechanical analysis of tunnels reinforced with various types of rock bolts (Osgoui and Oreste, 2010; Bobet and Einstein, 2011; He et al., 2015; Cai et al., 2015; Sun et al., 2021, 2023, 2024). These studies provide valuable insights into the characteristics of the GRCs and the evolution of the plastic zone. However, these solutions assume that the tunnel surrounding rock is in an axisymmetric stress regime, making it difficult to account for the true three-dimensional high tectonic stress environment in which tunnels exist. Advanced analytical methods that consider complex stress states must be developed to investigate the mechanical behavior of reinforced surrounding rock. The complex variable method has achieved success in the derivation of analytical solutions for the mechanics of surrounding rock in underground tunnels, demonstrating its unique advantages (Kargar et al., 2014; Lu et al., 2014; Zhang et al., 2022). Lu et al. (2018) established a two-dimensional elastic-plastic mechanical analysis model for unreinforced tunnel surrounding rock based on the complex variable method. An analytical method was presented for solving the stress distribution and plastic zone boundaries under non-hydrostatic stress conditions, demonstrating that the shape of the plastic zone is significantly influenced by the lateral pressure coefficient. Our recent research (Ma et al., 2024) has demonstrated that when a significant difference exists between gravitational stress and tectonic stress, a localized plastic zone may develop in the surrounding rock of the tunnel, with its extent gradually increasing as the geostress level increases. Furthermore, we found that the stress concentration factor K_{SCF} effectively captures the disparity between gravitational and tectonic stresses. The authors suggest that the development of a semi-analytical method for the mechanical analysis of tunnel surrounding rock under non-hydrostatic stress conditions,

accounting for the three-dimensional strength of rock masses, can effectively address the need for rapid stability analysis in high tectonic stress environments. However, existing methods are limited to homogeneous rock mass (i.e., where the surrounding rock parameters remain unchanged) and therefore fail to account for changes in mechanical properties and parameters caused by localized grouting reinforcement. This results in an insufficient ability to address the challenges associated with analyzing tunnel stability after grouting reinforcement and conducting sensitivity studies on reinforcement parameters. The applicability of existing methods must be further examined when optimizing gradient grouting parameters for the mechanical analysis of surrounding rock in functionally graded reinforced zones (FGRZs), as illustrated in Fig. 2).

Considering the highly concentrated stress distribution characteristics in the shallow depth of the surrounding rock after tunnel excavation, the roles of the natural rock zone and the reinforced zone in surrounding rock stability differ significantly. Using the same large-scale grouting reinforcement methods and parameters for both shallow and deep surrounding rock can lead to material waste and undermine the effective mobilization of the rock mass's self-supporting capabilities. Considering that the shallow depth of the surrounding rock is characterized by high stress concentration after tunnel excavation, the natural rock zone and the reinforced zone play different roles in surrounding rock stability. The application of a uniform grouting reinforcement method and parameters across extensive areas of the surrounding rock, particularly in both shallow and deep regions, results in material waste and undermines the optimal utilization of the self-supporting characteristics of the rock mass. In this context, the paper proposes a functional gradient bearing approach for the grouting reinforcement of tunnel surrounding rock. Initially, a mechanical analysis model is established that considers the combined effects of the natural rock zone and the reinforced zone. Subsequently, the complex variable method is employed to analyze the elastic-plastic mechanical response of both the natural rock zone and FGRZs under high tectonic stress conditions, as well as to determine the distribution of the plastic zone. This study proposes a semi-analytical method based on some essential assumptions during derivation. The application of the proposed method is subject to certain limitations, as comprehensively detailed in Section 2 and Section 7. The proposed method is further validated through comparative analysis with traditional solutions. The study provides a rapid assessment of the improvement effects of functionally graded reinforcement on the stress and bearing characteristics of tunnels at varying depths, providing new insights for the stress-controlled mechanism of deep-buried tunnels based on functional gradient reinforcement.

2. Problem description

2.1. Basic assumptions

Grouting reinforcement projects are often concealed and large-scale, and the geological conditions related to equipment, site, and in-situ rock mass at the construction location are complex. Therefore, significant challenges arise in evaluating the quality and determining the parameters of reinforced rock masses. Generally, numerical simulation experiments (Li et al., 2022; Yao et al., 2023) or on-site in-situ testing (Chen et al., 2015; Kim et al., 2019; Xie et al., 2021; Ma et al., 2023a) can be attempted to obtain the evaluation results. But the inconsistencies observed between results from these approaches raise significant concerns. Therefore, this study introduces the concept of an equivalent reinforced zone to describe the impact of advanced grouting on the natural rock zone during the mechanical analysis. Initially, it is assumed that the effects of advanced grouting reinforcement can be equivalently represented as an improvement in the mechanical properties of the surrounding rock. Based on this assumption, a semi-analytical method for elastoplastic mechanics is proposed to analyze the reinforcement effect in the natural rock zone and reinforced zones, under the condition that the mechanical parameters of the surrounding rock after grouting reinforcement are predefined. Subsequently, a parameter sensitivity analysis is conducted by selecting a reasonable range of reinforcement parameters, focusing primarily on the deformation and strength parameters that significantly affect the stability of the surrounding rock. However, the precise determination of the mechanical parameters under the discussed working conditions was not fully addressed. Instead, the parameters were assigned values within a reasonable range to the extent possible. The proposed method exhibits limitations in accurately obtaining the parameters of the rock masses in reinforced zones. The findings can provide a reference for the design of engineering grouting schemes and the construction of grouting reinforcement. The following assumptions are also made when developing the mechanical analysis model for the tunnel surrounding rock:

(1) It is assumed that the tunnel is sufficiently long, with zero strain along the tunnel axis, satisfying the plane strain condition at sections far from the tunnel face. Studies have shown that the tunnel face constraint effect dissipates at distances greater than three times the tunnel diameter. The assumption is acceptable and does not result in significant calculation errors (Chu et al., 2019, 2024; Xu et al., 2022).
(2) This study assumes a circular tunnel cross-section geometry combined with full-face grouting reinforcement. Nevertheless, this assumption makes it impossible to consider the influence of tunnel shape on the stress distribution in both the original rock zone and the reinforced zone, thereby making it difficult to accurately evaluate its effect on surrounding rock stability control. It should be noted that full-face grouting reinforcement is typically recommended only when encountering adverse geological conditions such as weak surrounding rock, high water pressure strata, or fault zones.
(3) The reinforced zone is further divided into two annular regions with distinct material properties. It is assumed that the reinforcement reduces the convergence deformation of the surrounding rock to a negligible level relative to the tunnel size, ensuring that the small deformation assumption holds throughout the elastic-plastic deformation stage.
(4) Both the natural rock mass and the grouted reinforced rock mass are assumed to be homogeneous and isotropic materials, with their stress-strain behavior described by an ideal elastic-plastic model (Fig. 1a). Research indicates that this model is suitable for poor-quality rock masses, while strain-softening models are more appropriate for medium-quality rock masses. Nevertheless, the ideal elastic-plastic model can still be used as an equivalent simulation for medium-quality rock masses, with the equivalent parameters determined by a forward analysis approach (Su et al., 2023).
(5) The strength characteristics of both unreinforced and reinforced rock masses are described by the generalized Zhang-Zhu (GZZ) criterion, and the enhancement of strength in the reinforced zone is equivalently represented as a modification of strength parameters. This approach is adopted solely for the purpose of facilitating parametric analysis. The accurate determination of reinforcement parameters is not addressed in this study. More details are further described in Section 2.2.

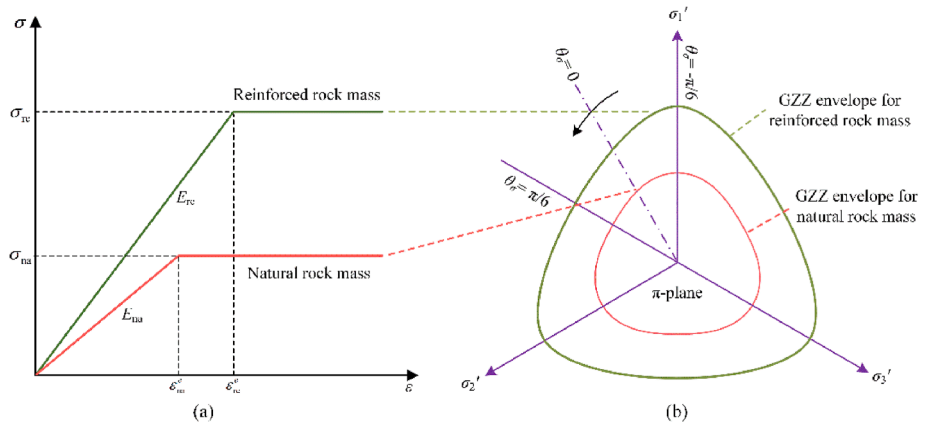


Fig. 1. Illustration of the mechanical characteristics of natural and reinforced rock masses: (a) the ideal elastoplastic stress-strain relationship, and (b) the GZZ strength envelopes for reinforced and natural rock masses on the π-plane.

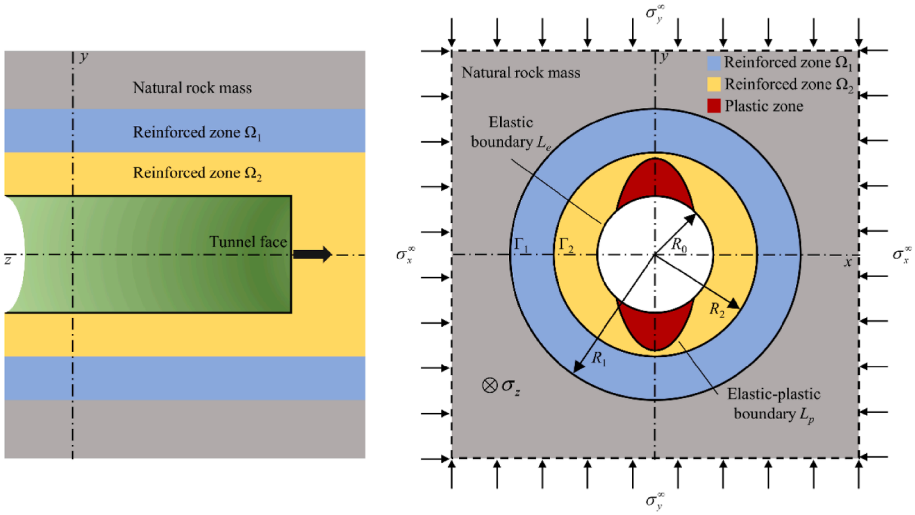


Fig. 2. Schematic representation of the mechanical model for a deep-buried circular tunnel with advanced full-face grouting under high tectonic stress and the development of complex-shaped plastic zones in the reinforced zone.

2.2. GZZ rock mass 3D strength theory

The success of stress-controlled disaster prevention and control of deep-buried tunnels relies on accurate stress analysis. The stress analysis requires the introduction of theoretical models applicable to deep rock masses and corresponding parameter systems with reliable sources to ensure the calculation accuracy. The mechanical parameters and theoretical models of rock masses hold equally important positions in research. The generalized Hoek-Brown (GHB) criterion, as an important representative of empirical strength criteria, effectively captures the nonlinear strength and geometric characteristics of joints in natural rock masses. The parameter determination methods associated with the criterion have been extensively validated in engineering practice, leading to their widespread application in rock engineering (Hoek and Brown, 2019).

However, in terms of theoretical models for rock masses, the GHB criterion ignores the significant influence of the intermediate principal stress in rock mass strength, making it difficult to apply in deep rock engineering under complex three-dimensional high-stress conditions (Trzeciak and Sone, 2022; Cai et al., 2023). The GZZ criterion, which fully inherits the parameter system of the GHB criterion, has been continuously revised and improved into a smooth, three-dimensional, and nonlinear extension of the GHB criterion, making it well-suited for deep rock engineering (Cai et al., 2021). In this paper, the smooth GZZ strength criterion is used to describe the failure behavior of both unreinforced and reinforced rock masses.

As illustrated in Fig. 1a, the stress-strain responses of both the natural and reinforced rock masses follow an ideal elastic-plastic model. Fig. 1b shows the relative positions of the strength envelopes for the natural and reinforced rock masses in the π-plane. The strength envelopes for the natural and reinforced rock masses are described by the GZZ strength criterion with different parameters. And the expression of GZZ is as follows:

$$F_{GZZ} = \frac{1}{\sigma_c^{(1/a-1)}}\left(\sqrt{3J_2}\right)^{1/a} + \left(\frac{3+\sin 3\theta_\sigma}{2\sqrt{3}}\right) m_b \sqrt{J_2} - m_b \frac{I_1}{3} - s\sigma_c = 0 \tag{1}$$

where σ_c is the uniaxial compressive strength (UCS) of intact rock, m_b, s, and a are empirical parameters that reflect the characteristics of rock quality and structure. The parameter θ_σ denotes the

stress Lode angle, which reflects the influence of the intermediate principal stress σ_2 on strength. Additionally, I_1 represents the first invariant of stress, and J_2 denotes the second invariant of the deviatoric stress. The expressions for θ_σ, I_1, and J_2 are as follows:

$$\left.\begin{aligned} \theta_\sigma &= \tan^{-1}\left[\frac{2\sigma_2-(\sigma_1+\sigma_3)}{\sqrt{3}(\sigma_1-\sigma_3)}\right] \\ I_1 &= \sigma_1+\sigma_2+\sigma_3 \\ J_2 &= \frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_1-\sigma_3)^2}{6} \end{aligned}\right\} \tag{2}$$

Hoek and Brown (2019) provided the calculation expressions for the strength parameters m_b, s, and a based on the Geological Strength Index (GSI) and blast damage index D:

$$\left.\begin{aligned} m_b &= m_i \exp[(GSI-100)/(28-14D)] \\ s &= \exp[(GSI-100)/(9-3D)] \\ a &= 1/2+[\exp(-GSI/15)-\exp(-20/3)]/6 \end{aligned}\right\} \tag{3}$$

where m_i is a material parameter that reflects the quality characteristics of the rock. Hoek and Brown (2019) recommend selecting this parameter from a table. In reinforced zones, grout enhances the strength by filling the joints within the rock mass. The parameters that characterize the intrinsic strength properties of intact rock remain essentially unchanged. It appears that the strength of reinforced rock masses can be approximated by increasing the GSI values. Although this approach may not be entirely consistent with engineering practices, it does not affect the conclusions regarding the effects of grouting reinforcement on tunnel surrounding rock stability analysis.

Recent studies indicate that this parameter is also related to the porosity of the rock, and the value of m_i can be modified based on experimental results (He et al., 2022). Conventional manual methods for obtaining the GSI inevitably involve subjectivity and require considerable labor and time. In contrast, the GZZ strength-based three-dimensional analysis theory utilizes a rock joint information investigation method based on digital in-situ testing (photogrammetry), allowing for the rapid acquisition of the GSI and real-time grading of rock engineering quality (Zhang et al., 2020, 2024b, 2024c). This advancement enables three-dimensional stress analysis and real-time design in deep rock engineering (Cai et al., 2022b).

2.3. Mechanical model

According to the research (Fang et al., 2013), the variations in the range and rock mass mechanical parameters of the reinforced zone under different engineering geological conditions may lead to the initiation of plastic zones either at the tunnel boundary or at the inner boundary of the reinforced zone. Under the same stress and reinforcement conditions, appropriate design of reinforcement and support parameters ensures that the plastic zone develops only within the reinforced zone, helping control rock mass deformation and enhancing surrounding rock stability. Moreover, for unreinforced fractured rock mass tunnels, the range of the plastic zone around the excavation is generally large and can completely encompass the entire circular tunnel. However, after advanced grouting reinforcement, the extent of the plastic zone will be reduced to some degree, potentially resulting in a plastic zone with non-closed boundaries. In our recent research (Ma et al., 2024), a general solution that considers the two types of plastic zones with different morphological characteristics was proposed. Through case analysis, it was found that the differences between gravitational stress and tectonic stress also significantly influence the morphology of the plastic zone. This paper will further investigate the morphological evolution of the plastic zone within the rock mass reinforcement zone. To maintain generality, the illustration and derivation will focus on plastic zones with non-closed boundaries.

As shown in Fig. 2, consider a long deep-buried circular tunnel with a radius of R_0. Advanced grouting establishes two reinforced zones with radii R_1 and R_2 ahead of the tunnel face. In this study, the shallow reinforced zone in the surrounding rock is denoted as Ω_1, while the deep reinforced zone is denoted as Ω_2. The mechanical properties of these two zones are purposefully differentiated to demonstrate the functional gradient reinforcement effect explored in this work. The surrounding rock is subjected to far-field horizontal (tectonic) stress σ_x^∞, vertical (gravitational) stress σ_y^∞, and axial stress σ_z perpendicular to the tunnel face. Under the three-dimensional high-stress conditions, a complex-shaped plastic zone forms within the reinforced zone. The plastic zone exhibits pronounced vertical expansion while maintaining limited horizontal extent when influenced by high tectonic stress. This paper illustrates the plastic zone with non-closed boundaries. Let the elastic boundary and the plastic boundary of the reinforced zone Ω_2 be L_e and L_p, respectively. When given geological conditions and reinforced zones, the morphological characteristics of the plastic zone can only be determined based on geostresses, surrounding rock parameters, and reinforcement parameters. This paper does not distinguish morphological characteristics of the plastic zone in the derivation. The derived solution method covers the case where the plastic zone completely surrounds the circular tunnel, providing a unified solution that accounts for the plastic zone of the surrounding rock under grouting reinforcement. The general solution approaches for these two plastic zone types and their morphological evolution are detailed in our previous work (Ma et al., 2024).

3. Mechanical analysis of natural rock zone and reinforced zone

According to the previous analysis, it is beneficial for the surrounding rock stability when the plastic zone only occurs within the reinforced zone Ω_2. Therefore, this paper conducts an in-depth study of this specific mode. In this situation, the surrounding rock in the reinforced zone Ω_2 will partially be in an elastic state, while both the reinforced zone Ω_1 and the natural rock zone will be entirely in an elastic state. The following sections present the mechanical analysis methods applicable to the elastic and plastic zones. For shallow-buried tunnels, the gravity effect of the surrounding rock becomes more significant than the geostresses, making it difficult to ignore in both the elastic and plastic zones (Kargar and Rahmannejad, 2015; Cai et al., 2020). However, this paper considers tunnels with sufficient burial depth, the gravity effect of the rock mass within the excavation disturbed zone can be neglected without introducing significant computational errors. This treatment has been used in many studies addressing the elastic-plastic problem of deep-buried tunnels (Guan et al., 2018, 2020; Ghorbani and Hasanzadehshooiili, 2019; Kargar, 2019).

3.1. Stress solution for plastic zone in reinforced rock mass

This paper assumes that the rock mass in the reinforced zone follows an ideal elastic-plastic model, where the post-peak strength is independent of plastic flow. During the small deformation stage, the changes in the shape and size of the tunnel boundary can be ignored. The stress distribution in the plastic zone is statically determined by the strength criteria and equilibrium

equations, representing a classical type of boundary value problem in plastic mechanics. This serves as the foundation for the development of slip line theory based on two-dimensional strength criteria in plastic mechanics (Jaeger et al., 2007). When considering the three-dimensional stress state, the displacement boundary condition along the tunnel axis supplements the relationship between the out-of-plane and in-plane stress components, ensuring that the problem remains statically determinate (Xu and Xia, 2021; Chen et al., 2022a; Cai et al., 2023). For such statically determinate elastic-plastic problems, far-field stresses affect only the shape and size of the plastic zone in the surrounding rock but do not influence the stress distribution in the plastic zone. Furthermore, when there are no external loads acting on the inner boundary of the circular tunnel, the plastic stress distribution is axial-symmetrical (Behnam et al., 2014; Lu et al., 2018). Chen et al. (2022a, b) extended traditional slip line theory to three-dimensional stress state based on the GZZ criterion, finding that three-dimensional strength significantly influences the distribution of the plastic stress field. This approach was then successfully applied to strip foundations. For the problem discussed in this paper, the plastic stress field can be quickly obtained using the differential method provided in the reference (Ma et al., 2024), given that the tunnel has a circular boundary. However, the method proposed by Chen et al. (2022a, b) is recommended for tunnels with complex shapes.

Early research primarily relied on two-dimensional strength theories, which exhibit significant limitations: (1) while the theory simplifies the computation process, it is unsuitable for deep-buried rock tunnels under high stress and a three-dimensional complex stress state; (2) the analytical process focuses only on in-plane mechanical responses, neglecting the impact of axial stress σ_z on the strength, deformation, and failure modes of the surrounding rock, whereas out-of-plane stress significantly affects in-plane stress and surrounding rock stability (Zhou and Li, 2011; Guan et al., 2018). In the elastic-plastic analytical calculations that consider the three-dimensional strength of the rock mass and three-dimensional stress state, both in-plane and out-of-plane stresses in the plastic zone may exhibit softening behavior. Cai et al. (2023) reviewed the method for determining out-of-plane stress and the relationship between in-plane and out-of-plane stresses. The following relationship is approximately satisfied between the axial stress σ_z in the plastic zone and the in-plane stress under the assumption of plane strain and within the framework of GZZ theory.

$$\sigma_z = \frac{1}{2}(\sigma_r + \sigma_\alpha) \tag{4}$$

where σ_r and σ_α represent the radial stress and circumferential stress in polar coordinates. This means that as the in-plane plastic strain increases, the out-of-plane stress decreases to the average of the other two principal stresses. This results in a discontinuity in the circumferential stress along the elastic-plastic boundary. But the magnitude of this discontinuity gradually vanishes as the Poisson's ratio μ of surrounding rock increases (Xu and Yu, 2006).

For the problem discussed in this paper, the in-plane stress components in plastic zones should satisfy the equilibrium differential equation:

$$\frac{d\sigma_r}{dr} - \frac{\sigma_\alpha - \sigma_r}{r} = 0 \tag{5}$$

Then, the plastic stress field can be determined from Eqs. (1), (4) and (5), with the detailed derivation process available in the literature (Ma et al., 2024).

3.2. Stress and displacement solution for elastic zone in natural/reinforced rock mass

In Fig. 2, the natural rock mass, the rock mass in reinforced zone Ω_1, and part of the rock mass in reinforced zone Ω_2 are all in an elastic state. Due to the different material properties of these three regions and the complex contact relationships between them, the stress distribution at the boundaries remains unknown. When employing inverse or semi-inverse methods from elasticity to solve such problems, the approach mainly relies on trial-and-error, making it generally difficult to directly obtain the solutions.

The complex variable method is particularly suitable for solving cases with complex geometric boundaries or load distributions. Its core involves conformal mapping to transform the original problem from the physical plane to the complex plane for analysis. Boundary conditions are then applied to obtain the solution. The problems discussed in this paper can be categorized into two types. The first type involves an infinite simply-connected domain with complex-shaped cavities, which can be mapped to the exterior (or interior) of a unit circle in the image plane, corresponding to the entire natural rock zone. The second type consists of an annular finite doubly-connected domain with complex inner and outer boundaries, which can be mapped to a simple annular region in the image plane (with the outer radius typically fixed at 1), corresponding to the elastic parts of the reinforced zones Ω_1 and Ω_2. It is worth noting that each type generally needs to be mapped to the image plane separately for discussion. This implies that both the mapping functions and the corresponding image planes differ across elastic regions. The general mapping process for these two types of problems is shown in Fig. 3.

According to the complex variable theory, the stress distribution in the elastic zone of the surrounding rock can be described by two analytic functions:

$$\sigma_x + \sigma_y = 4\mathrm{Re}\left[\varphi_1'(z)\right] \tag{6}$$

$$\sigma_y - \sigma_x + 2\mathrm{i}\tau_{xy} = 2\left[\overline{z}\varphi_1''(z) + \psi_1'(z)\right] \tag{7}$$

where σ_x and σ_y represent the normal stress components along the x and y directions in the Cartesian coordinate system, while τ_{xy} denotes the shear stress component. The symbol i denotes the imaginary unit. Re[] denotes the operation of taking the real part of the function within the bracket, $\overline{z}$ denotes the complex conjugate of z, and similar notation will not be explicitly explained further in the following. The degree of determination for $\varphi_1(z)$ and $\psi_1(z)$ varies depending on the discussed problem.

The complex variable representation of the displacements in the elastic zone is given by

$$2G(u + \mathrm{i}v) = \kappa\varphi_1(z) - z\overline{\varphi_1'(z)} - \overline{\psi_1(z)} \tag{8}$$

where u and v are the displacement along x and y directions, respectively. G represents the shear modulus, which is related to the elastic modulus E and Poisson's ratio μ by the formula $G = E/2(1+\mu)$. κ is dependent on μ, and for plane strain problems, we have $\kappa = 3 - 4\mu$.

The displacement boundary condition can be directly obtained by substituting the boundary values into Eq. (8). The stress boundary conditions expressed in terms of stress components are as follows:

$$\left.\begin{aligned} X_n &= \sigma_x l + \tau_{xy} m \\ Y_n &= \tau_{xy} l + \sigma_y m \end{aligned}\right\} \tag{9}$$

where X_n and Y_n are the specified surface force components along

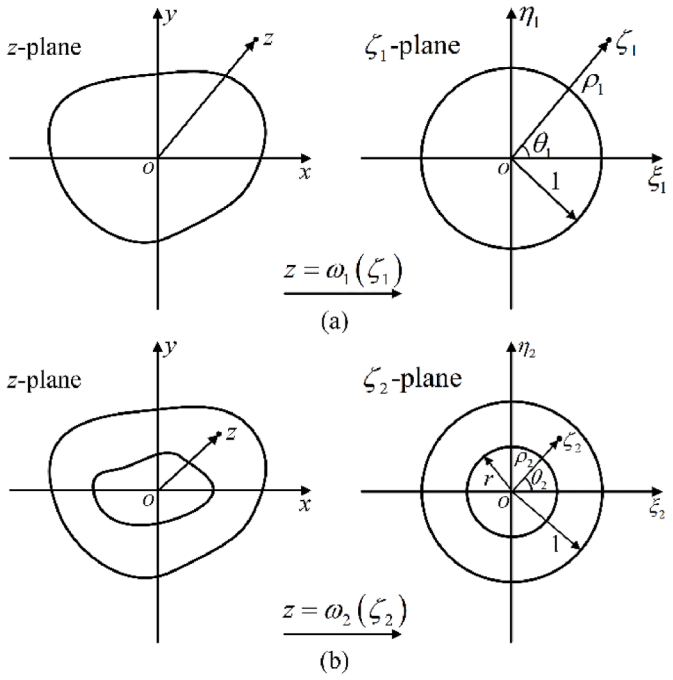


Fig. 3. Conformal mapping from the physical plane to the image plane: (a) an infinite simply-connected domain with complex inner boundaries mapped to the exterior of a unit circle, and (b) an annular doubly-connected domain with complex inner and outer boundaries mapped to a simple annular region.

the x-axis and y-axis, respectively. l and m are the direction cosines of the unit outward normal at the discussed boundary points, and $l = \cos(n,x)$, $m = \cos(n,y)$, n represents the direction of the outward normal at the boundary.

The representation of the stress boundary condition in complex variable form is

$$\varphi(z) + z\overline{\varphi'(z)} + \overline{\psi(z)} = \mathrm{i}\int_A^B (X_n + \mathrm{i}Y_n)ds + CI \tag{10}$$

where z should be understood as the boundary value, A is the starting point of integration on the boundary, and B is an arbitrary point on the boundary. When integrating from A to B, it is essential to ensure that the region under discussion lies to the left of the direction of advance. CI is a complex constant determined by point A, and it can sometimes be taken as zero.

4. Solution of plastic zone

4.1. Analysis of stress and displacement in natural rock zone

Let a point in the natural rock zone in the physical plane be z_1. To ensure generality, we consider the natural rock as an infinite domain in the physical plane (z_1-plane) with a complex inner boundary shape, which is mapped to the exterior of a circle with a radius r_1 of 1 in the image plane (ζ_1-plane). The general form of the mapping function is

$$z_1 = \omega_1(\zeta_1) = R\left(\zeta_1 + \sum_{k=0}^{\infty} C_k \zeta_1^{-k}\right) \tag{11}$$

where R and C_k are mapping function coefficients. When the natural rock zone is symmetric with respect to the x-axis, the mapping function coefficients are real numbers. For specific problems, a finite number of terms for C_k can yield satisfactory accuracy. When the inner boundary of the natural rock zone is circular, we have $R = R_1$, with the other coefficients equal to zero.

The surrounding rock within the natural rock zone is in an elastic state. The principle of elastic superposition is applicable. The stress and displacement fields within the natural rock zone, resulting from the combined influence of geostresses and the reinforced zones, can be effectively decoupled into two independent contributions: (1) the primary mechanical response induced solely by geostresses, and (2) the secondary perturbative effects attributable exclusively to the reinforced zones. When considering only the effects of geostresses, we can use two single-valued analytic functions $\varphi_{11}(z_1)$ and $\psi_{11}(z_1)$ to represent the stress and displacement in the natural rock zone. The general forms of $\varphi_{11}(z_1)$ and $\psi_{11}(z_1)$ in the infinite domain are

$$\varphi_{11}(z_1) = \Gamma z_1 + \sum_{k=0}^{\infty} a'_k z_1^{-k} \tag{12}$$

$$\psi_{11}(z_1) = \Gamma' z_1 + \sum_{k=0}^{\infty} b'_k z_1^{-k} \tag{13}$$

where $\Gamma = \left(\sigma_x^{\infty} + \sigma_y^{\infty}\right)/4$, $\Gamma' = \left(\sigma_y^{\infty} - \sigma_x^{\infty}\right)/2$, a'_k and b'_k are the unknown coefficients of the analytic functions.

By substituting Eq. (11) into Eqs. (12) and (13) respectively, we obtain the two analytic functions expressed with ζ_1 as the variable in the exterior of the unit circle:

$$\varphi_{11}(\zeta_1) = \varphi_{11}(z_1) = \Gamma\omega_1(\zeta_1) + \sum_{k=0}^{\infty} a_k \zeta_1^{-k} \tag{14}$$

$$\psi_{11}(\zeta_1) = \psi_{11}(z_1) = \Gamma'\omega_1(\zeta_1) + \sum_{k=0}^{\infty} b_k \zeta_1^{-k} \tag{15}$$

where a_k and b_k are the unknown coefficients of the analytic functions, and they are the quantities to be determined in this paper. The coefficients have been extensively studied by numerous scholars and can be determined using methods such as Cauchy integral and power series (e.g., Kargar et al., 2014; Zhao and Yang, 2015; Setiawan and Zimmerman, 2020).

Since Eq. (11) is also an analytic function, the analytic function originally represented by the complex variable z_1 transforms into an analytic function represented by the complex variable ζ_1 after conformal mapping. The analytic function expressed in terms of ζ_1 exhibits improved properties.

Similarly, when considering only the effects of the reinforced zones, we can represent the stress and displacement in the natural rock zone using two analytic functions $\varphi_{12}(\zeta_1)$ and $\psi_{12}(\zeta_1)$ expressed with ζ_1 as the variable in the exterior of the unit circle. The functions are

$$\varphi_{12}(\zeta_1) = c_0 + \sum_{k=1}^{\infty} c_k \zeta_1^{-k} \tag{16}$$

$$\psi_{12}(\zeta_1) = d_0 + \sum_{k=1}^{\infty} d_k \zeta_1^{-k} \tag{17}$$

where c_0, d_0, c_k, and d_k are the unknown coefficients of the analytic functions, representing the quantities to be determined in this paper.

Based on Eqs. (6) and (7), when considering the combined effects of geostresses and the reinforced zones, the stress in the natural rock zone can be expressed as

$$\sigma_x+\sigma_y=4\mathrm{Re}[\Phi_1(\zeta_1)] \tag{18}$$

$$\sigma_y-\sigma_x+2\mathrm{i}\tau_{xy}=2\left[\frac{\overline{\omega_1(\zeta_1)}}{\omega_1(\zeta_1)}\frac{\Phi_1'(\zeta_1)}{\omega_1'(\zeta_1)}+\Psi_1(\zeta_1)\right] \tag{19}$$

where $\Phi_1(\zeta_1)=\varphi_1'(\zeta_1)/\omega_1'(\zeta_1)$, $\varphi_1(\zeta_1)=\varphi_{11}(\zeta_1)+\varphi_{12}(\zeta_1)$, $\Psi_1(\zeta_1)=\psi_1'(\zeta_1)/\omega_1'(\zeta_1)$, $\psi_1(\zeta_1)=\psi_{11}(\zeta_1)+\psi_{12}(\zeta_1)$, $\varphi_1(\zeta_1)$ and $\psi_1(\zeta_1)$ are the two analytic functions representing the real stresses in the natural rock zone.

According to Eq. (8), the displacement in the natural rock zone when only considering the effects of geostresses is

$$2G_1(u_{11}+\mathrm{i}v_{11})=\kappa_1\varphi_{11}(\zeta_1)-\frac{\omega_1(\zeta_1)}{\omega_1'(\zeta_1)}\overline{\varphi_{11}'(\zeta_1)}-\overline{\psi_{11}(\zeta_1)} \tag{20}$$

where u_{11} and v_{11} are the displacement along x and y directions, respectively. G_1 represents the shear modulus of natural rock mass, and κ_1 is a material constant that depends on the Poisson's ratio μ_1 of natural rock mass.

According to Eq. (8), when considering only the effects of the reinforced zones, the presence of the reinforced zone not only causes a stress redistribution in the natural rock zone but also restricts some displacements of the surrounding rock. The displacements of the surrounding rock limited by the reinforced zone can be expressed using the analytic functions $\varphi_{12}(\zeta_1)$ and $\psi_{12}(\zeta_1)$ as follows:

$$2G_1(u_{12}+\mathrm{i}v_{12})=\kappa_1\varphi_{12}(\zeta_1)-\frac{\omega_1(\zeta_1)}{\omega_1'(\zeta_1)}\overline{\varphi_{12}'(\zeta_1)}-\overline{\psi_{12}(\zeta_1)} \tag{21}$$

where u_{12} and v_{12} are the displacement along x and y directions, respectively.

Therefore, when considering the interaction between the natural rock zone and the reinforced zone, the displacement in the natural rock zone is

$$2G_1(u+\mathrm{i}v)=\kappa_1\varphi_1(\zeta_1)-\frac{\omega_1(\zeta_1)}{\omega_1'(\zeta_1)}\overline{\varphi_1'(\zeta_1)}-\overline{\psi_1(\zeta_1)} \tag{22}$$

where $u+\mathrm{i}v=u_{11}+u_{12}+\mathrm{i}(v_{11}+v_{12})$.

4.2. Analysis of stress and displacement in the reinforced zones

Let the points within the reinforced zone Ω_1 and the reinforced zone Ω_2 in the physical plane be z_2 and z_3, respectively. To ensure generality in our derivation, we consider the reinforced zone Ω_1 as an annular domain with both complex inner and outer boundaries, mapping it to a simple annular region in the ζ_2-plane, with a fixed outer radius of 1 and an inner radius of r_2. The general form of the mapping function is

$$z_2=\omega_2(\zeta_2)=\sum_{k=-\infty}^{\infty}D_k\zeta_2^k \tag{23}$$

where D_k denotes the mapping function coefficient. When the reinforced zone is symmetric with respect to the x-axis, the mapping function coefficients are real numbers. For specific problems, a finite number of terms for D_k can achieve satisfactory accuracy. When the reinforced zone is an annulus, we have $D_k=R_2$ and $r_2=R_1/R_2$.

The elastic part of the reinforced zone Ω_2 has Γ_1 as the outer boundary, and L_e and L_p as the inner boundaries (Fig. 2). It is mapped to a simple annular region in the ζ_3-plane, with a fixed outer radius of 1 and an inner radius of r_3. The general form of the mapping function is identical to that in Eq. (23). We denote it as

$$z_3=\omega_3(\zeta_3)=\sum_{k=-\infty}^{\infty}E_k\zeta_3^k \tag{24}$$

In this paper, the mapping function coefficients R and C_k in Eq. (11) are determined from the coordinates of its inner boundary, while the mapping function coefficients D_k in Eq. (23) are obtained based on the coordinates of both its inner and outer boundaries. Although the determination of the coefficients is a fundamental step in the method employed in this paper, it is not the focus or a novel contribution of our research. Readers interested in the methods for determining such mapping function coefficients are referred to the work by Ma et al. (2022), which provides a review and introduces a general method for finding the mapping functions. However, since the plastic boundary is unknown, the mapping function coefficients E_k in Eq. (24) and the inner radius r_3 of the annulus remain undetermined. The mapping coefficients are the key unknowns to be solved in this study. The solution of the coefficients will be detailed in Section 4.5.

The reinforced zone Ω_1 and elastic parts of the reinforced zone Ω_2 are annular finite doubly-connected domains. According to the complex variable theory, the two analytic functions $\varphi_2(z_2)$ and $\psi_2(z_2)$ in the finite domain (reinforced zone Ω_1) should have the following general forms:

$$\varphi_2(z_2)=\sum_{k=-\infty}^{\infty}e_k'z_2^{-k} \tag{25}$$

$$\psi_2(z_2)=\sum_{k=-\infty}^{\infty}f_k'z_2^{-k} \tag{26}$$

where e_k' and f_k' are the unknown coefficients of the analytic functions.

By substituting Eq. (23) into Eqs. (25) and (26) respectively, we obtain the two analytic functions $\varphi_2(\zeta_2)$ and $\psi_2(\zeta_2)$ expressed with ζ_2 as the variable in the simple annular domain as follows:

$$\varphi_2(z_2)=\varphi_2(\zeta_2)=\sum_{k=1}^{\infty}e_{-k}\zeta_2^{-k}+e_0+\sum_{k=1}^{\infty}e_k\zeta_2^k \tag{27}$$

$$\psi_2(z_2)=\psi_2(\zeta_2)=\sum_{k=1}^{\infty}f_{-k}\zeta_2^{-k}+f_0+\sum_{k=1}^{\infty}f_k\zeta_2^k \tag{28}$$

where e_{-k}, e_0, e_k, f_{-k}, f_0, and f_k are the unknown coefficients of the analytic functions, representing the quantities to be determined in this paper.

Similarly, the stress and displacement in the elastic part of the reinforced zone Ω_2 can be expressed using two analytic functions $\varphi_3(\zeta_3)$ and $\psi_3(\zeta_3)$ in the simple annular domain with ζ_3 as the variable, as follows:

$$\varphi_3(\zeta_3)=\sum_{k=1}^{\infty}g_{-k}\zeta_3^{-k}+g_0+\sum_{k=1}^{\infty}g_k\zeta_3^k \tag{29}$$

$$\psi_3(\zeta_3)=\sum_{k=1}^{\infty}h_{-k}\zeta_3^{-k}+h_0+\sum_{k=1}^{\infty}h_k\zeta_3^k \tag{30}$$

where g_{-k}, g_0, g_k, h_{-k}, h_0, and h_k are the unknown coefficients of the analytic functions, representing the quantities to be determined in this paper.

According to Eqs. (6) and (7), the stresses in the reinforced zone Ω_1 are

$$\sigma_x+\sigma_y=4\mathrm{Re}[\Phi_2(\zeta_2)] \tag{31}$$

$$\sigma_y-\sigma_x+2\mathrm{i}\tau_{xy}=2\left[\frac{\overline{\omega_2(\zeta_2)}}{\omega_2'(\zeta_2)}\Phi_2'(\zeta_2)+\Psi_2(\zeta_2)\right] \tag{32}$$

where $\Phi_2(\zeta_2)=\varphi_2'(\zeta_2)/\omega_2'(\zeta_2)$, $\Psi_2(\zeta_2)=\psi_2'(\zeta_2)/\omega_2'(\zeta_2)$.

According to Eqs. (6) and (7), the stresses in the elastic part of the reinforced zone Ω_2 are

$$\sigma_x+\sigma_y=4\mathrm{Re}[\Phi_3(\zeta_3)] \tag{33}$$

$$\sigma_y-\sigma_x+2\mathrm{i}\tau_{xy}=2\left[\frac{\overline{\omega_3(\zeta_3)}}{\omega_3'(\zeta_3)}\Phi_3'(\zeta_3)+\Psi_3(\zeta_3)\right] \tag{34}$$

where $\Phi_3(\zeta_3)=\varphi_3'(\zeta_3)/\omega_3'(\zeta_3)$, $\Psi_3(\zeta_3)=\psi_3'(\zeta_3)/\omega_3'(\zeta_3)$.

According to Eq. (8), the displacements in the reinforced zone Ω_1 are

$$2G_2(u+\mathrm{i}v)=\kappa_2\varphi_2(\zeta_2)-\frac{\omega_2(\zeta_2)}{\overline{\omega_2'(\zeta_2)}}\overline{\varphi_2'(\zeta_2)}-\overline{\psi_2(\zeta_2)} \tag{35}$$

where G_2 is the shear modulus of the reinforced zone Ω_1, and κ_2 is a material constant that depends on the Poisson's ratio μ_2 of the reinforced zone Ω_1.

According to Eq. (8), the displacements in elastic part of the reinforced zone Ω_2 are

$$2G_3(u+\mathrm{i}v)=\kappa_3\varphi_3(\zeta_3)-\frac{\omega_3(\zeta_3)}{\overline{\omega_3'(\zeta_3)}}\overline{\varphi_3'(\zeta_3)}-\overline{\psi_3(\zeta_3)} \tag{36}$$

where G_3 is the shear modulus of the reinforced zone Ω_2, and κ_3 is a material constant that depends on the Poisson's ratio μ_3 of the reinforced zone Ω_2.

4.3. Boundary conditions for solving analytic functions

When considering only the effects of geostresses, the coefficients in the analytic functions $\varphi_{11}(\zeta_1)$ and $\psi_{11}(\zeta_1)$ can be determined by the stress boundary conditions at the inner boundary of the natural rock zone. According to Eqs. (10) and (11), the stress boundary conditions at the inner boundary of the natural rock zone are

$$\varphi_{11}(t_1)+\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_{11}'(t_1)}+\overline{\psi_{11}(t_1)}=\mathrm{i}\oint(X_n+\mathrm{i}Y_n)ds+CI_1 \tag{37}$$

where t_1 represents a point on the unit circle. $\varphi_{11}'(t_1)$ denotes the first derivative with respect to $\varphi_{11}(t_1)$, and this notation will not be illustrated in the following. CI_1 is an undetermined constant and can be set to 0. There are no external forces acting on the inner boundary of the natural rock zone when considering only the effects of geostresses. And thus, both X_n and Y_n are 0, making the right-hand side of Eq. (37) equal to 0. Based on Eqs. (11) and (37), the coefficients a_k and b_k in the analytic functions $\varphi_{11}(\zeta_1)$ and $\psi_{11}(\zeta_1)$ can be solved, with the detailed process provided in Appendix A.

On the boundary Γ_2 between the natural rock zone and the reinforced zone Ω_1, the stress continuity equation can be obtained from Eqs. (10), (11) and (23) as follows:

$$\varphi_1(t_1)+\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_1'(t_1)}+\overline{\psi_1(t_1)}=\varphi_2(t_2)+\frac{\omega_2(t_2)}{\overline{\omega_2'(t_2)}}\overline{\varphi_2'(t_2)}+\overline{\psi_2(t_2)}+CI_2 \tag{38}$$

where t_2 represents a point on the unit circle, and CI_2 is an undetermined integration constant.

Since $\varphi_{11}(\zeta_1)$ and $\psi_{11}(\zeta_1)$ have been provided in Appendix A, substituting Eq. (37) into Eq. (38) yields the stress continuity equation with $\varphi_{12}(\zeta_2)$, $\psi_{12}(\zeta_2)$, $\varphi_2(\zeta_2)$, and $\psi_2(\zeta_2)$ as unknown functions:

$$\varphi_{12}(t_1)+\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_{12}'(t_1)}+\overline{\psi_{12}(t_1)}=\varphi_2(t_2)+\frac{\omega_2(t_2)}{\overline{\omega_2'(t_2)}}\overline{\varphi_2'(t_2)}+\overline{\psi_2(t_2)}+CI_2 \tag{39}$$

According to Eqs. (8), (22) and (35), the displacement continuity equation at the boundary Γ_1 between the natural rock zone and the reinforced zone Ω_1 can be expressed as follows:

$$\frac{1}{G_1}\left[\kappa_1\varphi_1(t_1)-\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_1'(t_1)}-\overline{\psi_1(t_1)}\right]=\frac{1}{G_2}\left[\kappa_2\varphi_2(t_2)-\frac{\omega_2(t_2)}{\overline{\omega_2'(t_2)}}\overline{\varphi_2'(t_2)}-\overline{\psi_2(t_2)}\right] \tag{40}$$

By substituting Eqs. (20) and (21) into Eq. (40), the displacement continuity equation with $\varphi_{12}(\zeta_2)$, $\psi_{12}(\zeta_2)$, $\varphi_2(\zeta_2)$, and $\psi_2(\zeta_2)$ as unknown functions is obtained as follows:

$$-\frac{1}{G_1}\left[\kappa_1\varphi_{12}(t_1)-\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_{12}'(t_1)}-\overline{\psi_{12}(t_1)}\right]+\frac{1}{G_2}\left[\kappa_2\varphi_2(t_2)-\frac{\omega_2(t_2)}{\overline{\omega_2'(t_2)}}\overline{\varphi_2'(t_2)}-\overline{\psi_2(t_2)}\right]=\frac{1}{G_1}\left[\kappa_1\varphi_{11}(t_1)-\frac{\omega_1(t_1)}{\overline{\omega_1'(t_1)}}\overline{\varphi_{11}'(t_1)}-\overline{\psi_{11}(t_1)}\right] \tag{41}$$

Similarly, at the boundary Γ_2 between the reinforced zones Ω_1 and Ω_2, the stress continuity equation can be expressed based on Eqs. (10), (23) and (24) as follows:

$$\varphi_2(r_2t_2)+\frac{\omega_2(r_2t_2)}{\overline{\omega_2'(r_2t_2)}}\overline{\varphi_2'(r_2t_2)}+\overline{\psi_2(r_2t_2)}=\varphi_3(t_3)+\frac{\omega_3(t_3)}{\overline{\omega_3'(t_3)}}\overline{\varphi_3'(t_3)}+\overline{\psi_3(t_3)}+CI_3 \tag{42}$$

where t_3 represents a point on the unit circle, and CI_3 is an undetermined integration constant.

Based on Eqs. (35) and (36), the displacement continuity equation at the boundary Γ_2 between the reinforced zones Ω_1 and Ω_2 is given by

$$\frac{1}{G_2}\left[\kappa_2\varphi_2(r_2t_2)-\frac{\omega_2(r_2t_2)}{\overline{\omega_2'(r_2t_2)}}\overline{\varphi_2'(r_2t_2)}-\overline{\psi_2(r_2t_2)}\right]=\frac{1}{G_3}\left[\kappa_3\varphi_3(t_3)-\frac{\omega_3(t_3)}{\overline{\omega_3'(t_3)}}\overline{\varphi_3'(t_3)}-\overline{\psi_3(t_3)}\right] \tag{43}$$

Based on Eqs. (10) and (24), the stress boundary equation at the elastic inner boundary L ($L = L_e \cup L_p$) of the reinforced zone Ω_2 is given by

$$\varphi_3(r_3t_3)+\frac{\omega_3(r_3t_3)}{\overline{\omega_3'(r_3t_3)}}\overline{\varphi_3'(r_3t_3)}+\overline{\psi_3(r_3t_3)}=\mathrm{i}\oint(X_n+\mathrm{i}Y_n)ds+CI_4 \tag{44}$$

where CI_4 is an undetermined constant and can be set to 0. When the stress distribution in the plastic zone and the stress boundary condition at the tunnel inner boundary of the tunnel are known, X_n and Y_n can be determined by Eq. (9), and readers are encouraged to refer to our previous work (Ma et al., 2024) for more details.

4.4. Method for solving the analytic function coefficients

In this paper, there are 6 unknown analytic functions $\varphi_{12}(\zeta_2)$, $\psi_{12}(\zeta_2)$, $\varphi_2(\zeta_2)$, $\psi_2(\zeta_2)$, $\varphi_3(\zeta_3)$, $\psi_3(\zeta_3)$. In practical calculation, the number of terms in the analytic functions expressed as infinite series can only take finite values. In this paper, both the highest positive and negative powers are set to n, resulting in a total of $10n+6$ terms. The terms are c_0, c_k, d_0, d_k, e_0, e_{-k}, e_k, f_0, f_{-k}, f_k, g_0, g_{-k}, g_k, h_0, h_{-k}, h_k ($k = 1, 2, \ldots, n$). Based on Eqs. (39) and (41)-(44), a total of $10n+5$ equations can be formulated regarding the unknowns. According to the Saint-Venant principle, the influence of the reinforced zone on the natural rock zone decreases gradually from shallow to deep layers until it vanishes. That is, as $\zeta_1 \to \infty$, the displacements u_{12} and v_{12} in the natural rock zone tend toward 0. We have the following relationship according to Eq. (21):

$$\lim_{\zeta\to\infty}[2G_1(u_{12}+\mathrm{i}v_{12})]=0 \tag{45}$$

And we obtain

$$d_0=\kappa_1 c_0 \tag{46}$$

The key step in solving the mechanical response of surrounding rock using the complex variable method is the determination of the analytic functions. To solve for these functions, boundary conditions must be applied, along with Eqs. (39) and (41)–(44) and (46), yielding a system of $10n+6$ equations. Theoretically, by solving this system of equations, the $10n+6$ analytic function coefficients can be uniquely determined. The power series method, Cauchy integral method, and analytic continuation method are three conventional approaches for solving analytic functions, among which the power series method is the most widely applicable and convenient. However, the equations for solving the analytic functions contain three mapping functions for the problem addressed in this paper. This implies that, in general, the same physical points on the boundaries Γ_1 or Γ_2 correspond to different image points in the image plane. More precisely, although the variables t_1, t_2, and t_3 in the equations are all located on the unit circle, the polar angles corresponding to these variables are different. The Cauchy integral method and analytic continuation method are no longer applicable for such problems. And the traditional power series method, which establishes a system of equations for determining the coefficients of the analytic functions by comparing the coefficients of the same powers of the same independent variable on both sides of the equations (Lu et al., 2014; Lyu et al., 2020; Zhang et al., 2022; Lv et al., 2023), is also difficult to apply for solving the analytic functions expressed in terms of three different independent variables.

Recently, Zhou et al. (2023) proposed a method that combines the boundary collocation method with the power series method to address the problem of determining the coefficients of analytic functions with different independent variables, successfully solving the stress analysis problem in orthotropic rock masses. We introduce the boundary collocation method to address the problem in this study, which can reduce computational effort and accommodate complex boundary conditions. In the calculations, a point z_1 is selected on the boundary Γ_1 between the natural rock zone and reinforced zone Ω_1 in the z-plane, which corresponds to a point t_1 in the ζ_1-plane. Meanwhile, a point z_2 is selected on the outer boundary Γ_1 of the reinforced zone Ω_1 in the z-plane, corresponding to a point t_2 in the ζ_2-plane. According to the mapping functions (11) and (23), we obtain

$$z_1=\omega_1(t_1)=R\left(t_1+\sum_{k=0}^{\infty}C_k t_1^{-k}\right) \tag{47}$$

$$z_2=\omega_2(t_2)=\sum_{k=-\infty}^{\infty}D_k t_2^k \tag{48}$$

Let z_1 and z_2 denote the same point in the z-plane, that is, $z_1 = z_2$. Substituting this into Eqs. (47) and (48) yields

$$R\left(t_1+\sum_{k=0}^{\infty}C_k t_1^{-k}\right)=\sum_{k=-\infty}^{\infty}D_k t_2^k \tag{49}$$

When the coefficients of the mapping functions in Eq. (49) are known, we can first assign a value to t_1 and then solve a univariate polynomial equation to obtain t_2. Since the mapping functions are analytic, there will always be a unique t_2 that lies on the unit circle. Similarly, once t_2 is determined, we let z_2 and z_3 represent the same point in the z-plane. According to Eqs. (23) and (24), the relationship between t_1 and t_2 can be obtained as follows:

$$\sum_{k=-\infty}^{\infty}D_k t_2^k=\sum_{k=-\infty}^{\infty}E_k t_3^k \tag{50}$$

When the mapping function coefficients in Eq. (50) are known, specifying t_2 allows for the unique determination of t_3, ensuring it lies on the unit circle. Subsequently, the analytic functions $\varphi_{12}(\zeta_2)$, $\psi_{12}(\zeta_2)$, $\varphi_2(\zeta_2)$, $\psi_2(\zeta_2)$, $\varphi_3(\zeta_3)$, and $\psi_3(\zeta_3)$ can be obtained by solving Eqs. (39) and (41)–(44) and (46).

4.5. Solution of plastic zone

The GZZ failure surface in the principal stress space and its strength envelope on the π-plane are illustrated in Fig. 4. It can be observed that the GZZ criterion accurately captures the nonlinear tension-compression transformation law of strength on the π-plane, as well as the nonlinear brittle-ductile failure characteristics on the meridian plane. To reflect the influence of the stress state and stress level on the yield or failure of the rock mass, a dimensionless failure approaching index FAI_{GZZ} is introduced to quantitatively evaluate the yield or failure degree of the rock element. FAI_{GZZ} describes the relative position of a point P in the principal stress space concerning the strength envelope, specifically revealing the relationship between the stress state at point P and the ultimate stress state. A FAI_{GZZ} value closer to 1 indicates an increased proximity to failure (Cai et al., 2022c; Ma et al., 2024).

When point P lies within the smooth GZZ envelope, the rock mass remains in an elastic state, and FAI_{GZZ} is defined as

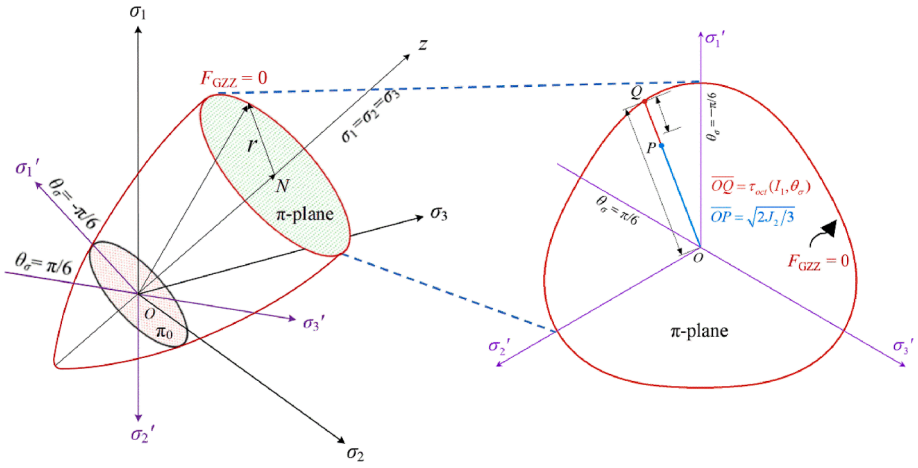


Fig. 4. Stress state of the rock and the smooth GZZ strength envelope in the π-plane (Ma et al., 2024).

$$FAI_{GZZ}=\frac{\overline{OP}}{\overline{OQ}}=\frac{\tau_{oct}(\sigma_1,\sigma_2,\sigma_2)}{\tau_{oct}(I_1,\theta_\sigma)}=\frac{\sqrt{2J_2/3}}{\tau_{oct}(I_1,\theta_\sigma)}=\frac{\sqrt{\left[(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_1-\sigma_3)^2\right]}}{3\tau_{oct}(I_1,\theta_\sigma)} \tag{51}$$

where $\overline{OQ}=\tau_{oct}(I_1,\theta_\sigma)$ represents the radius of the smooth GZZ strength envelope in the π-plane, which can be obtained by solving the GZZ strength criterion (Eq. (1)). When $FAI_{GZZ}=1$, point P coincides with point Q, indicating that the rock is in a critical failure state. When the rock is under hydrostatic pressure conditions ($\sigma_1=\sigma_2=\sigma_3$), corresponding to the origin O, $FAI_{GZZ}=0$, signifying that the rock experiences no yield or failure. Hence, we have

$$\left.\begin{array}{l} FAI_{GZZ}(I_1,J_2,\theta_\sigma)=1 \ \text{ In plastic state} \\ FAI_{GZZ}(I_1,J_2,\theta_\sigma)<1 \ \text{ In elastic state} \end{array}\right\} \tag{52}$$

The boundary of the plastic zone is determined by the limit equilibrium condition of the surrounding rock. Inside the boundary, the rock is in a plastic state, while outside, it remains elastic. At the elastic-plastic interface, where points P and Q coincide, the failure approaching index FAI_{GZZ} reaches 1. When the plastic zone forms only within a localized area of the reinforced zone Ω_2, the rock elements on the elastic inner boundary of the tunnel will reach their maximum load-bearing capacity, with the FAI_{GZZ} value as close to 1 as possible. Whether the plastic zone develops only in a localized area or completely surrounds the circular tunnel, the outer boundary of the elastic zone in the reinforced region Ω_2 is fixed at Γ_2, while the inner boundary never encroaches into the free surface of the tunnel. This paper considers the outer boundary of the reinforced zone Ω_2 to be circular. The geometric conditions that the elastic part of it must satisfy can be described as follows:

$$\left.\begin{array}{l} r(z_3)=\sqrt{\omega_3(t_3)\overline{\omega_3(t_3)}}=R_2 \\ r(z_3)=\sqrt{\omega_3(r_3t_3)\overline{\omega_3(r_3t_3)}}>=R_0 \end{array}\right\} \tag{53}$$

The optimization techniques will be employed to solve this type of elastic-plastic problem in this study. The mapping function coefficients in Eq. (24), which describe the inner and outer boundaries of the elastic part of the reinforced zone Ω_2, will be treated as design variables X. The objective is to find an optimal solution X^* that satisfies the following conditions. Let an arbitrary point on the unit circle in the ζ_3-plane be denoted as t_{3j}, corresponding to a point z_{3j} in the elastic zone of the reinforced region Ω_2. The mathematical model for solving this type of problem can be expressed as follows:

$$\left.\begin{array}{l} \min\left[f(X)=\sum f_j(X)\right], z_{3j}=\omega_3\left(r_3t_{3j}\right)\in L_p \\ \text{s.t. } g_j(X)<=0, z_{3j}=\omega_3\left(r_3t_{3j}\right)\in L \\ h_j(X)=0, z_{3j}=\omega_3\left(t_{3j}\right)\in \Gamma_2 \end{array}\right\} \tag{54}$$

where $f(X)=\sum f_j(X)$, $f_j(X)=\left[1-FAI_{GZZ}(I_1,J_2,\theta_\sigma)|_{z_{3j}}\right]^2$, with a minimum value of 0. $f_j(X)$ represents the square of the difference between FAI_{GZZ} and 1 at any point on the elastoplastic interface L_p. And $g_j(X)=R_0-|z_{3j}|$, $h_j(X)=|z_{3j}|-R_2$.

Using the penalty function method, $g_j(X)$ and $h_j(X)$ can be added as penalty terms to the objective function $f(X)$. And then, a new unconstrained augmented objective function $F(X)$ is constructed:

$$F(X)=f(X)+r_1\sum\left(\max\left[0,g_j(X)\right]\right)^2+r_2\sum\left(\min\left[0,h_j(X)\right]\right)^2 \tag{55}$$

where r_1 and r_2 are the penalty factors for $g_j(X)$ and $h_j(X)$, typically chosen to be relatively large numbers during calculations. It has been found that the Differential Evolution algorithm or the Particle Swarm Optimization algorithm can effectively yield optimal results.

5. Method verification

To validate the correctness of the theoretical derivation, we compare the proposed method with two existing analytical methods. This includes the analysis of elastic-plastic mechanical responses under different strength criteria for both axisymmetric and non-axisymmetric stress distributions. Subsequently, an indirect verification is conducted by examining the contact conditions at the boundaries of the reinforced zones and the elastic-plastic boundaries.

5.1. Mechanical analysis of axisymmetric stress distribution model

Fang et al. (2013) proposed an analytical solution for the mechanical responses of surrounding rock under full-section grouting reinforcement conditions based on the Mohr-Coulomb criterion. The correctness of their solution has been validated by both

classical analytical methods and numerical simulations. The mechanical model involved in this method can be reduced to the model considered by Fang et al. To facilitate validation, the parameters of the reinforced zone Ω_1 are set identical to those of the natural rock zone, with rock masses governed by the Mohr-Coulomb strength criterion. The tunnel radius R_0 is set to 5 m, with both horizontal and vertical stresses equal to 4 MPa. Other relevant parameters are shown in Table 1. Although it may seem necessary to adopt the same parameters as Fang et al. (2013) for comparative validation, it is important to recognize that such mechanical parameter selections for the reinforced zone may not be reasonable in practical engineering. This is due to the complex influence of factors such as grouting pressure, equipment specifications, and construction techniques on the mechanical properties of rock masses after grouting reinforcement. Accurately determining these parameters remains a significant challenge in underground engineering, as discussed in Section 2.1.

Fig. 5 presents the comparison results of the two methods, including the radius of the plastic zone for the reinforced zone Ω_2 at radii R_2 of $2R_0$, $2.5R_0$, $3R_0$, $3.5R_0$, and $4R_0$, as well as the radial stress σ_r and tangential stress σ_α on the inner side of boundary Γ_2^- and outer side of boundary Γ_2^+ of the reinforced zone. The plastic zone radius and stress distribution of the surrounding rock obtained from the two methods are very close, indicating the correctness of the model and analytical method presented in this paper. It can also be concluded that the degree of stress concentration $(\sigma_\alpha - \sigma_r)$ in the reinforced zone is higher than that in the natural rock zone. Specifically, after reinforcement, the stress concentration in the surrounding rock is transferred to the shallow surrounding rock (near the tunnel wall). However, the tunnel does not experience significant damage under high concentrated stresses due to the enhanced load-bearing capacity of the reinforced rock. The expansion of the grouting range also restricts the development of the plastic zone into the deeper surrounding rock, thereby enhancing the stability of the surrounding rock. However, once the reinforcement range surpasses a certain threshold, the effectiveness of this improvement becomes limited. Therefore, indiscriminately increasing the size of the reinforced zone is not cost-effective.

5.2. Mechanical analysis of non-axisymmetric stress distribution model

Lu et al. (2018) developed a Hoek-Brown-based elastoplastic method for complex geostress conditions, which assume that the surrounding rock of the tunnel is a single homogeneous medium, failing to consider the non-uniform parameter distribution caused by reinforcement. This method can be employed to determine the distribution of the plastic zone in both unreinforced and fully reinforced cases. By establishing identical parameters for both the reinforced zone and the natural rock zone, a comparative validation of the two analytical methods can be performed. Fig. 6 illustrates the geometric characteristics of the complex elastoplastic boundary shapes obtained from both the proposed method and traditional methods under two different conditions. It also highlights the distribution characteristics of the plastic zone resulting from the functional gradient reinforcement effect in the surrounding rock regions as identified by the proposed method. In the validation, the increase in UCS is utilized to reflect the enhancement of rock strength due to reinforcement. The horizontal and vertical stresses are set to 40 MPa and 50 MPa, respectively, with the remaining mechanical parameters presented in Table 2.

Table 1
Mechanical parameters of the natural rock zone and reinforced zone when compared to the axisymmetric model.

Mechanical parameter	Natural rock zone	Reinforced zone
Cohesion, c (MPa)	0.6	1.2
Internal friction angle, φ (°)	10	20
Elastic modulus, E (GPa)	10	30
Poisson's Ratio, μ	0.45	0.35

Fig. 6 illustrates the shape and size of the plastic zones obtained by both methods, with the plastic zone radius normalized to the tunnel radius R_0. It also presents the characteristics of the plastic zone shape obtained using the proposed method for a reinforced zone with a radius of $2R_0$. Fig. 6 shows that, for both unreinforced and fully reinforced conditions, the shape and size of the plastic zones obtained from the proposed method and the traditional method are consistent, indicating the reliability of the proposed method. The traditional method, which can be employed for the mechanical analysis of tunnel surrounding rock subjected to non-hydrostatic geostresses under both unreinforced and fully reinforced conditions, is not suitable for analyzing the mechanical characteristics of annular partitioned reinforced rock. As shown in Fig. 6, significant differences are observed in the plastic zone shape of partitioned reinforced rock obtained using the proposed method, compared to those of unreinforced and fully reinforced rock. This highlights the limitations of traditional methods in analyzing the characteristics of functionally graded reinforced rock masses, while the method proposed in this paper provides a more accurate representation of the radial gradient reinforcement effect in circular tunnels subjected to non-hydrostatic geostresses.

5.3. Indirect verification of the proposed method

Based on the computational model presented in this paper, the degree of satisfaction of Eqs. (39) and (41)-(44) can be verified by examining the contact conditions at the boundaries of the reinforced zones Ω_1 and Ω_2 and the plastic zone boundary, thereby providing an indirect validation of the proposed method. The displacement in the rectangular coordinate system can be obtained using Eqs. (22), (35) and (36) when the analytic functions are given. And the stress can be calculated using Eqs. (18), (19) and (31)-(34). For comparison, stresses in the rectangular coordinate system are transformed into the orthogonal curvilinear coordinate system using the following transformation formulas:

$$\sigma_\rho + \sigma_\theta = \sigma_x + \sigma_y \tag{56}$$

$$\sigma_\theta - \sigma_\rho + 2\mathrm{i}\tau_{\rho\theta} = \left(\sigma_y - \sigma_x + 2\mathrm{i}\tau_{xy}\right)e^{2\mathrm{i}\theta} \tag{57}$$

where σ_ρ, σ_θ, $\tau_{\rho\theta}$ represent the normal stress, tangential stress, and shear stress in the orthogonal curvilinear coordinate system, respectively. And $e^{2\mathrm{i}\theta} = \frac{\zeta_j}{\rho_j^2}\frac{\omega_j'(\zeta_j)}{\overline{\omega_j'(\zeta_j)}}$, with $j = 1, 2, 3$ corresponding to the physical quantities in the natural rock zone, reinforced zone Ω_1, and reinforced zone Ω_2.

Consider a deep-buried circular tunnel in a high tectonic stress region, where the horizontal geostress σ_x^∞ is 40 MPa and the vertical geostress σ_y^∞ is 25 MPa. The radii of the reinforced zones Ω_1 and Ω_2 are 2.5 R_0 and 4 R_0, respectively. The effect of blasting disturbance is not considered, and the remaining mechanical parameters are shown in Table 3. When using the GZZ strength criterion for rock mass, the reinforcement effect of grouting is reflected by increasing the *GSI*. This paper transforms the inverse problem of finding the unknown elastoplastic interface into a

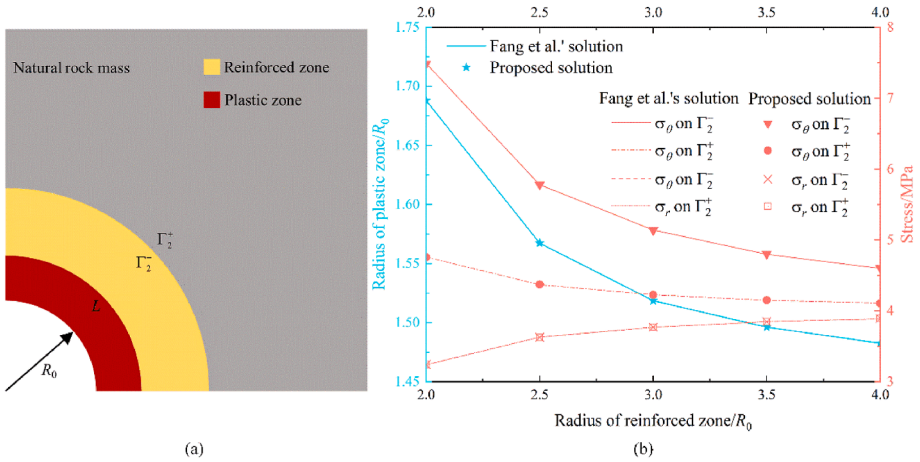


Fig. 5. Comparison of the method presented in this paper with the traditional method: (a) schematic of the reinforced zone and the natural rock zone, and (b) results of the radius of the plastic zone and the stress distribution on the boundary of the reinforced zone.

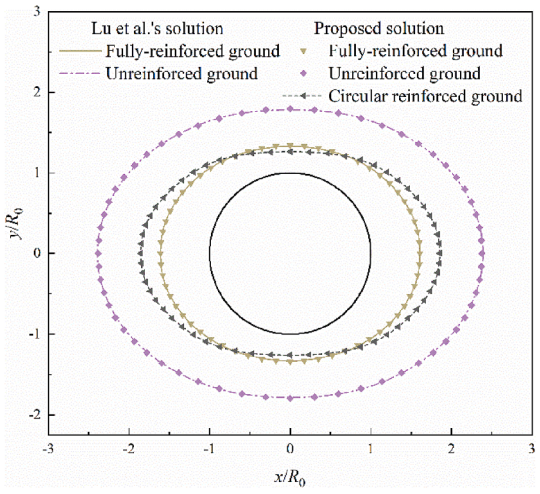


Fig. 6. Distribution patterns of plastic zones around circular tunnels subjected to non-hydrostatic geostresses obtained by the proposed method and traditional method.

problem of solving for mapping function coefficients. It demonstrates that both the Differential Evolution algorithm and Particle Swarm Optimization algorithm can effectively optimize the solution from the perspective of forward analysis.

To ensure computational accuracy, it is recommended that the mapping function includes more than 9 positive and negative power terms, while the analytical functions should exceed 20. The mapping function coefficients E_k in Eq. (24), obtained by the proposed method, are given in Table 4. The corresponding inner radius r_3 used in the mapping is 0.61668. Fig. 7 illustrates the elastoplastic interface and the mapped boundary Γ_2 of the reinforced zone derived from these coefficients. Fig. 8 shows the corresponding stress and displacement components along the reinforced zone boundaries Γ_1, Γ_2, and the elastoplastic interface L. The stress is normalized with respect to UCS σ_c of intact rock, and the displacement is normalized with respect to the tunnel radius R_0.

Due to the symmetry between the elastoplastic interface L and

Table 2
Mechanical parameters of the natural rock zone and the reinforced zone when compared to the non-axisymmetric model.

Mechanical parameter	Natural rock zone	Reinforced zone
UCS, σ_c (MPa)	10	20
Strength parameter, m_i	10	10
Strength parameter, s	0.5	0.5
Strength parameter, a	1	1
Elastic modulus, E (GPa)	10	30
Poisson's Ratio, μ	0.45	0.35

Table 4
The mapping function coefficients E_k in Eq. (24) obtained by the proposed method.

Mapping function coefficients, E_k	Values
E_{11}	1.2681×10^{-3}
E_9	2.8127×10^{-3}
E_7	8.0642×10^{-3}
E_5	4.682×10^{-3}
E_3	2.8668×10^{-3}
E_1	2.467
E_{-1}	-2.9421×10^{-1}
E_{-3}	-1.4268×10^{-2}
E_{-5}	-1.6334×10^{-3}
E_{-7}	-1.4965×10^{-4}
E_{-9}	1.0021×10^{-5}

Table 3
Mechanical parameters of the natural rock zone and reinforced zones of deep-buried tunnels in high tectonic stress zones.

Mechanical parameter	Natural rock zone	Reinforced zone Ω_1	Reinforced zone Ω_2
UCS, σ_c (MPa)	20	20	20
Strength parameter, m_i	20	20	20
GSI	20	40	60
Elastic modulus, E (GPa)	15	20	25
Poisson's Ratio, μ	0.45	0.4	0.35

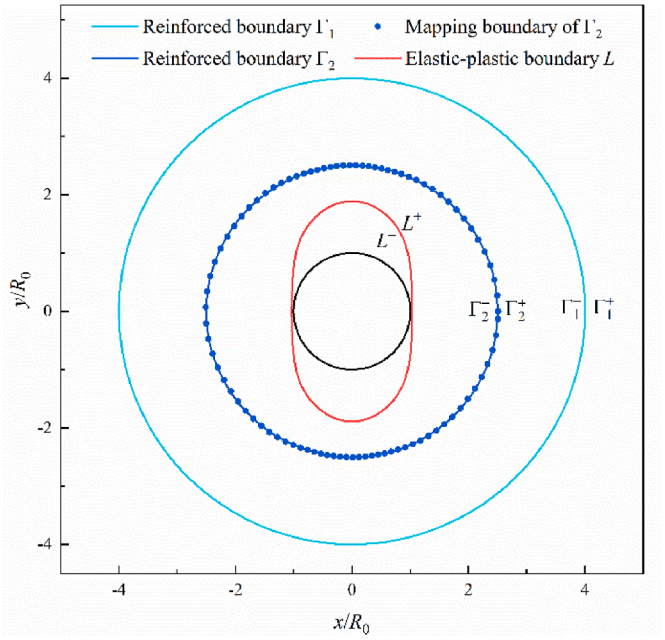


Fig. 7. The actual boundary of the reinforced zone on the physical plane, along with the mapped boundary and the elastoplastic interface obtained using the proposed method.

the boundary of the reinforced zone Γ_2, the mapping function coefficients E_k for even values of k are not included in Table 4. As shown in Fig. 7, the outer boundary of the annular region plotted by the mapping function, fits the actual boundary of the reinforced zone Γ_2 very well, indicating that the constraint $h_j(X)$ in the objective Eq. (55) is satisfied. Fig. 8 reveals that the normal stress, shear stress, and displacements along the reinforced zone boundaries Γ_1, Γ_2, and the elastoplastic boundary L are continuous, thereby satisfying the contact conditions. Additionally, the difference in mechanical parameters between the natural rock zone and the reinforced zones leads to discontinuities in the tangential stress components along Γ_1 and Γ_2. Therefore, the reliability of the proposed method is indirectly validated.

6. Example analysis

The morphological characteristics and evolution patterns of the plastic zone in surrounding rock around a circular tunnel are influenced by various factors. The results are presented in a normalized format.

6.1. Influence of tectonic stress on variable burial depth tunnels

Consider a deep-buried circular tunnel in high tectonic stress zones with a radius of R_0 and an overlying rock mass density of 25 kN/m^2. The plastic zones of the tunnel are obtained using the proposed method at burial depths of 600 m, 800 m, 1000 m, and 1200 m under varying horizontal/tectonic stress conditions. Within the framework of the fundamental assumptions and mechanical model, the evolution of the plastic zone around a circular tunnel under functionally gradient reinforcement can be systematically analyzed. As depicted in Fig. 9, the morphology of the plastic zone is significantly affected by burial depth and tectonic stress. The parameters for the natural rock zone and reinforced zone are kept consistent with those in Section 5.3.

Fig. 9 shows that there are significant differences in the plastic

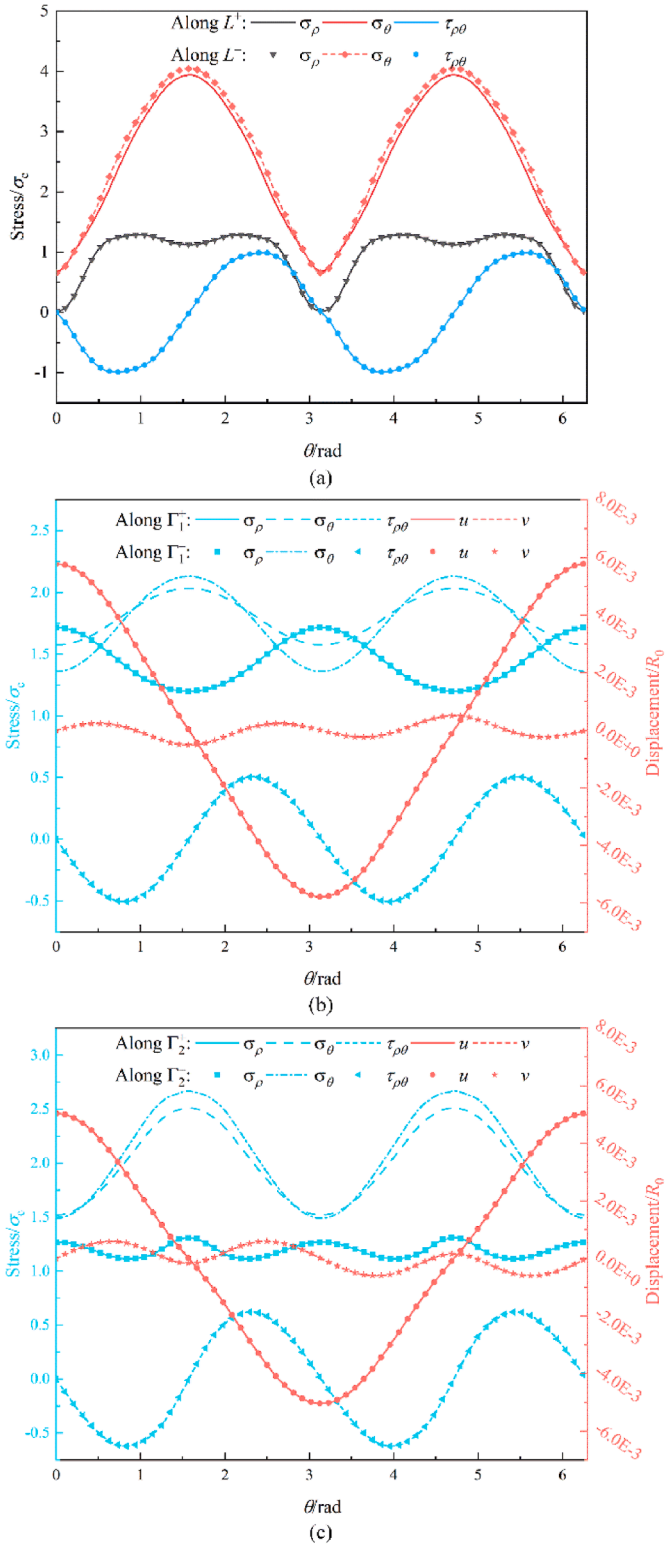


Fig. 8. Stress and displacement distributions along the reinforced zone boundaries and the elastoplastic interface: (a) stress and displacement components on both sides of boundary Γ_1, (b) stress and displacement components on both sides of boundary Γ_2, and (c) stress components on both sides of elastoplastic interface L.

zone morphology of the tunnel under a high tectonic stress field compared to a hydrostatic stress field, regardless of the burial depth of the tunnel. Under a constant lateral pressure coefficient,

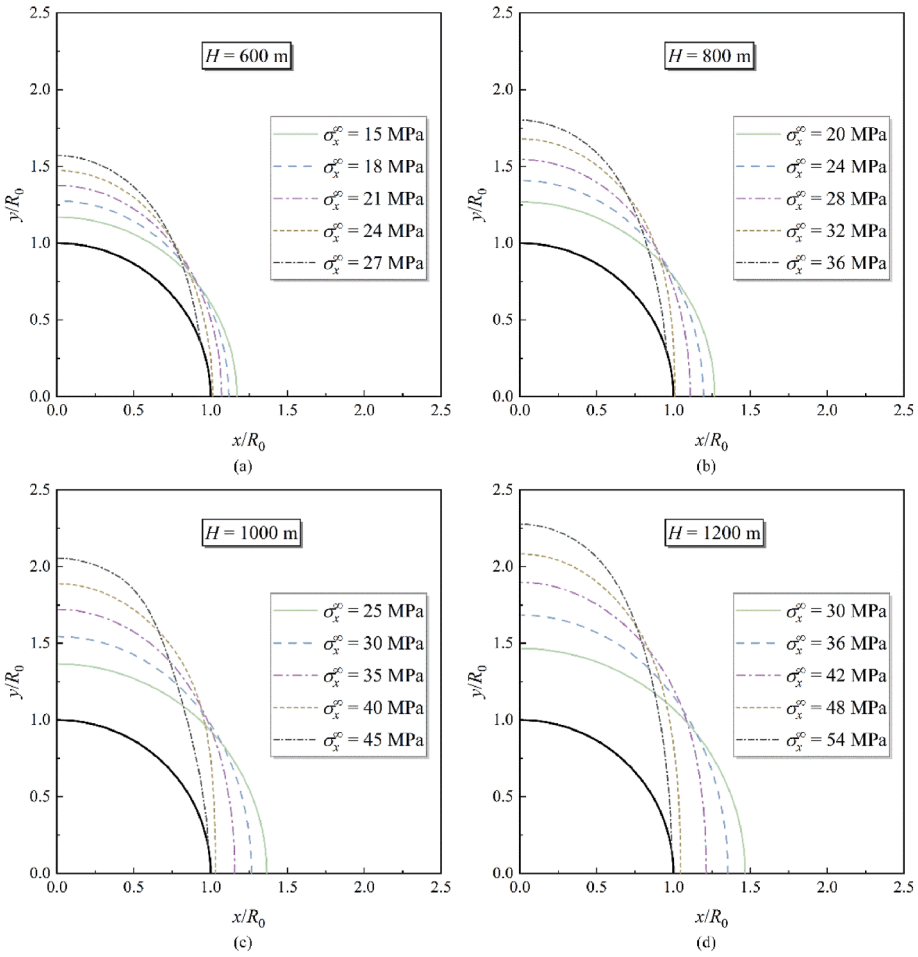


Fig. 9. The shape and size of the plastic zones in surrounding rock under different tectonic stresses: (a) H = 600 m, (b) H = 800 m, (c) H = 1000 m, and (d) H = 1200 m.

the plastic zone gradually extends deeper into the surrounding rock as the burial depth increases. The plastic zone at the tunnel crown and bottom expands as the tectonic stress increases at a constant burial depth, potentially leading to sustained plastic deformation and significant failure of the surrounding rock. The development of the plastic zone in the horizontal direction at the tunnel arch lumbar decreases. Although the proportion of elastic rock elements increases, the tangential stress and axial stress become closer under this stress state, bringing the rock elements closer to a triaxial tensile stress condition ($\sigma_1 = \sigma_2$), which can easily lead to tensile failure of the surrounding rock. When tectonic stress is excessively high, the surrounding rock may experience tensile stress in the horizontal direction. The GZZ criterion is established based on experimental data for rocks subjected to triaxial compression. Therefore, this paper indicates that the yielding and failure of the surrounding rock are driven by triaxial compressive stress when solving for the plastic stress field. The proposed method may no longer be applicable in extreme cases where a tensile-compressive stress state occurs in the surrounding rock. The proposed method predicts that the complex stress distribution in the surrounding rock may lead to mixed failure in extreme tectonic stress environments. The failure mode of surrounding rock in tunnels in high tectonic stress regions is more complex, accompanied by a decrease in the surrounding rock stability.

The authors (Ma et al., 2024) proposed the use of two dimensionless indices K_σ and K_{SCF} to evaluate the impact of high tectonic stress on the stability of deep-buried circular tunnels. K_σ and K_{SCF} are expressed as follows:

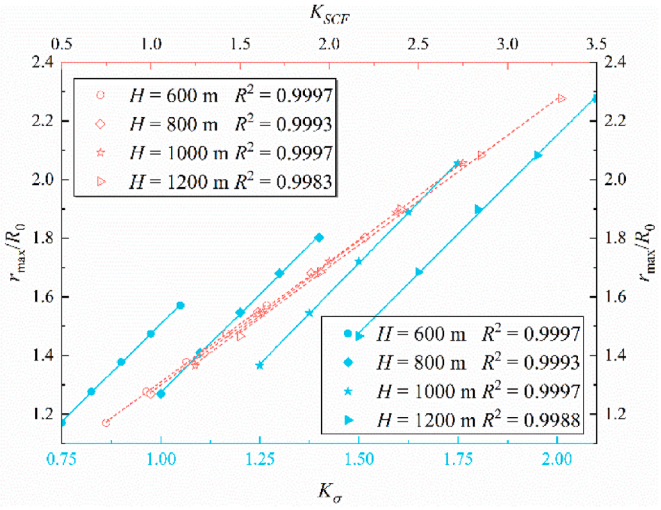


Fig. 10. The relationship between the maximum radius of the plastic zone at different burial depths and the two dimensionless factors K_σ and K_{SCF}.

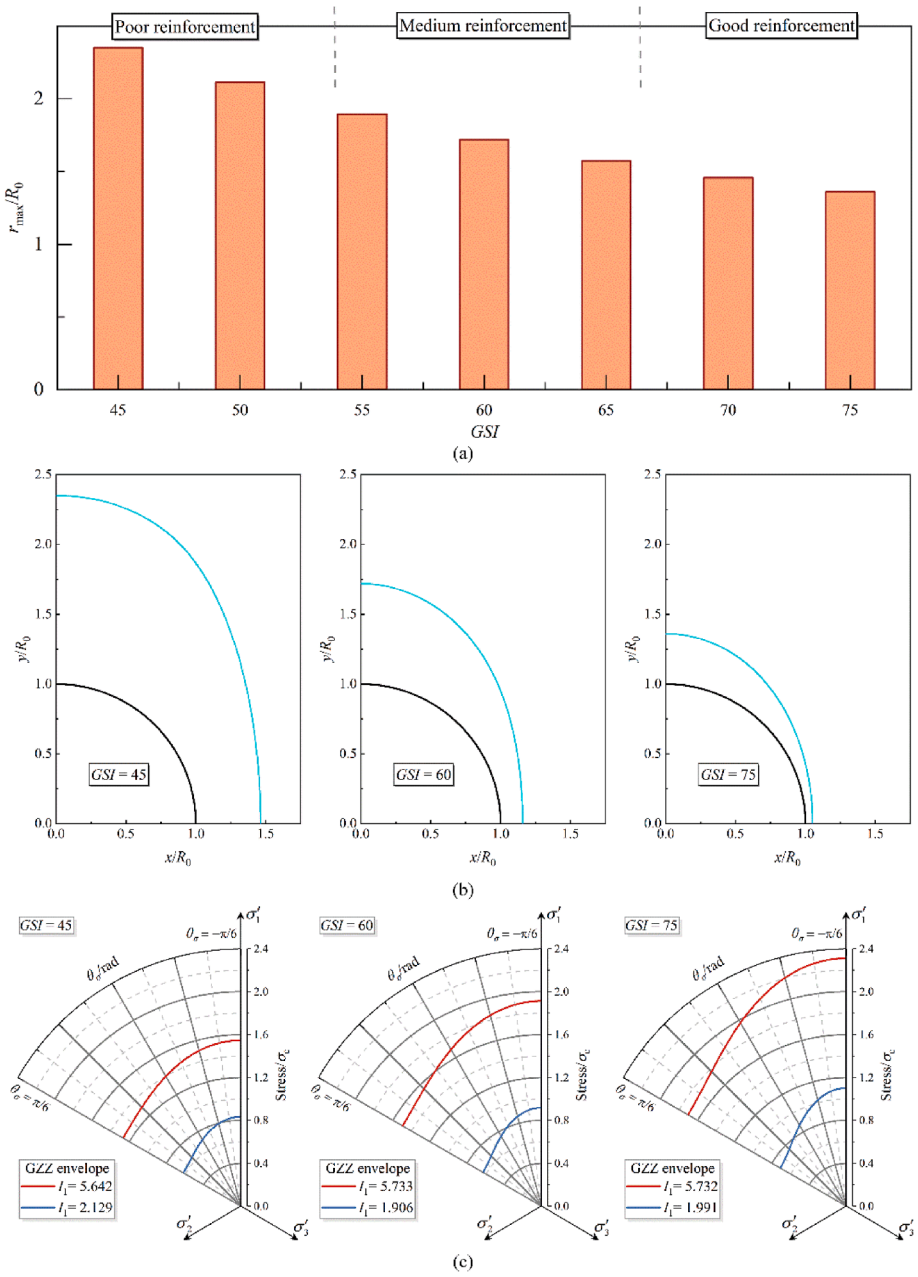


Fig. 11. Results for the surrounding rock in the reinforced zone Ω_2 under different reinforcement effects (*GSI* values): (a) maximum radius of the plastic zone, (b) evolution of the shape and size of the plastic zone, and (c) maximum and minimum GZZ envelopes corresponding to the stress state at the plastic zone boundaries.

$$\left.\begin{aligned} K_\sigma &= \frac{\sigma_x^\infty + \sigma_y^\infty}{2\sigma_c} \\ K_{SCF} &= \frac{3\sigma_x^\infty - \sigma_y^\infty}{2\sigma_c} \end{aligned}\right\} \tag{58}$$

where K_σ represents the ratio of the average value of horizontal and vertical stresses to the UCS of the intact rock, and K_{SCF} indicates the ratio of the maximum tangential stress at the surface of the circular tunnel to the UCS of the intact rock.

Let the maximum depth of the plastic zone be denoted as r_{max}. It was found by Ma et al. (2023b) that the maximum depth of the plastic zone increases approximately linearly with burial depth, based on a substantial amount of field measurement data. A linear empirical fitting formula relating r_{max} to H was then provided. It can be observed from Fig. 10 that this study theoretically further reveals the linear relationship. Even with reinforcement of the surrounding rock, the maximum radius r_{max} of the plastic zone exhibits a strong linear correlation with K_σ or K_{SCF}, and this relationship remains consistent across different burial depths. As a result, the approach of using K_σ and K_{SCF} to evaluate the maximum extent of the plastic zone in the surrounding rock of circular tunnels and to guide tunnel support design is valid.

6.2. Influence of GSI in the reinforced zone

The effect of grouting reinforcement on improving the bearing capacity of surrounding rock lacks quantitative assessment. This section investigates the inhibitory effect of grouting reinforcement on the development of the plastic zone by varying the *GSI* of the

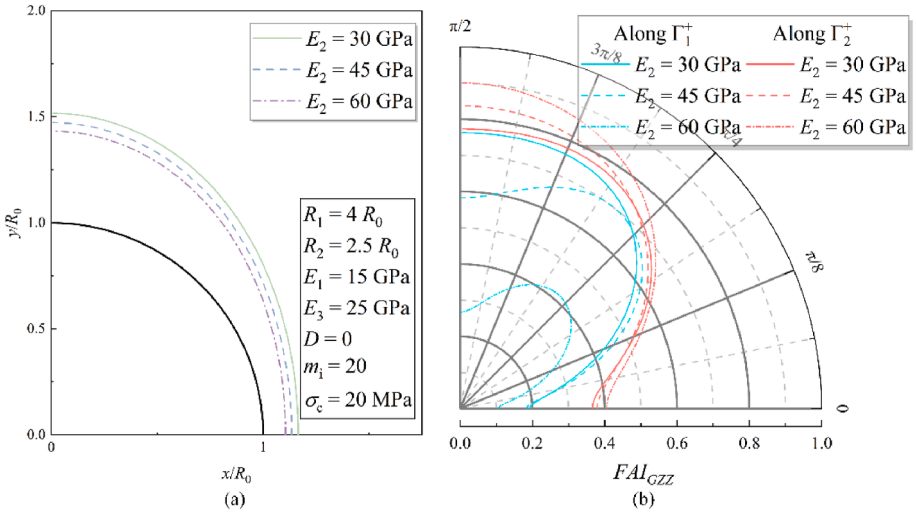


Fig. 12. Mechanical characteristics of surrounding rock in reinforced zone Ω_1 for different elastic moduli E_2: (a) shape and size of the plastic zones, and (b) failure approaching index FAI_{GZZ} at the reinforced zone boundaries.

reinforced rock mass. The variations in the shape and size of the plastic zone are explored for rock masses in the reinforced zone Ω_2 under different strengths within the specified grouting range, as shown in Fig. 11. The calculations were performed with the vertical stress σ_y^∞ fixed at 25 MPa and the horizontal stress σ_x^∞ at 35 MPa, while the other parameters remained consistent with those in Section 6.1.

Fig. 11a and b indicates that when the *GSI* value of the reinforced zone exceeds 45, the plastic zone develops in the reinforced zone Ω_2, which has a positive effect on limiting surrounding rock deformation and enhancing its load-bearing capacity. As the *GSI* value of the reinforced zone increases, the radius of the plastic zone at the tunnel crown decreases rapidly, while that at the arch lumbar of the tunnel decreases more slowly. When the *GSI* value exceeds 60, further improvements in grouting quality have a minimal impact on suppressing the development of the plastic zone. Fig. 11c presents the maximum and minimum GZZ envelope radii corresponding to the yielding of rock elements at the plastic zone boundary, defined by the stress states at the intersections of the plastic zone boundary with the *y*-axis and *x*-axis. The stress levels I_1 corresponding to the maximum GZZ envelope remain nearly identical under varying degrees of reinforcement. However, the stress states and strength characteristics exhibit significant differences. Under poor reinforcement ($GSI = 45$), the stress values along the maximum GZZ envelope are relatively low, indicating a lower degree of stress concentration. Conversely, under good reinforcement ($GSI = 75$), the stress values along the GZZ envelope are significantly higher, resulting in greater stress concentration. That is, a good reinforcement effect enables the surrounding rock to endure higher stress concentrations, resulting in failure. In summary, grouting reinforcement significantly enhances the stability of weakly fractured rock masses with low *GSI* values. But it is not recommended to control the plastic zone of the surrounding rock through excessive grouting.

6.3. Influence of the elastic modulus of surrounding rock in the reinforced zone Ω_1

The deformation and failure characteristics of the surrounding rock are influenced by both strength and stiffness. The above analysis focused on the impact of strength parameters in the reinforced zones on surrounding rock stability. Grouting reinforcement also modifies the deformation parameters of the surrounding rock, particularly the elastic modulus and Poisson's ratio. However, field measurements indicate that the Poisson's ratio exhibits no significant variation. Therefore, this study focuses only on the sensitivity analysis of the elastic modulus. The impact of the elastic modulus of the surrounding rock in the reinforced zone Ω_1 on the stress distribution and plastic zone extent is explored. In the calculations, the *GSI* of the surrounding rock in reinforced zone Ω_2 is set to 60, while other parameters remain the same as those in Section 6.2. The distribution patterns of the plastic zone and the failure approaching indices of the surrounding rock in reinforced zone Ω_1 are calculated for different values of E_2 (see Fig. 12). The stress distribution in the representative rock elements at the boundaries of the reinforced zones and the natural rock zone is illustrated in the π-plane (see Fig. 13), based on 6 observation points selected by rotating counterclockwise from the horizontal direction to the vertical direction.

As shown in Fig. 12a, increasing the elastic modulus E_2 of reinforced zone Ω_1 results in a uniform reduction of the plastic zone extent, which is beneficial for the stability of the surrounding rock. It can be seen from Fig. 12b that the failure approaching index FAI_{GZZ} on the boundary Γ_1^+ of the natural rock zone gradually decreases as E_2 increases, while the FAI_{GZZ} on the boundary Γ_2^+ in the reinforced zone Ω_1 gradually increases as E_2 increases. This indicates that increasing the rock stiffness in reinforced zone Ω_1 can effectively transfer the stress concentration area of the surrounding rock from the shallow zone to the deeper reinforced zone, thereby fully utilizing the stiffness and load-bearing potential of the reinforced surrounding rock.

The stress transfer law of the surrounding rock is further revealed in Fig. 13 as E_2 increases. The stress level I_1 in the natural rock zone remains nearly constant for different E_2 values, indicating that the reinforced zones nearly encompass the disturbance zone caused by tunnel excavation. The stress level and concentration degree at the boundary of reinforced zone Ω_1 significantly increase, requiring higher strength to ensure the stability of the reinforced rock mass. Therefore, the *GSI* of the surrounding rock in reinforced zone Ω_1 is set to 70 in Fig. 13f. The stress Lode angle θ_σ in the rock elements at all observation points is less than 0. This indicates a true triaxial compression state, which positively contributes to the potential of the compressive strength of reinforced rock masses.

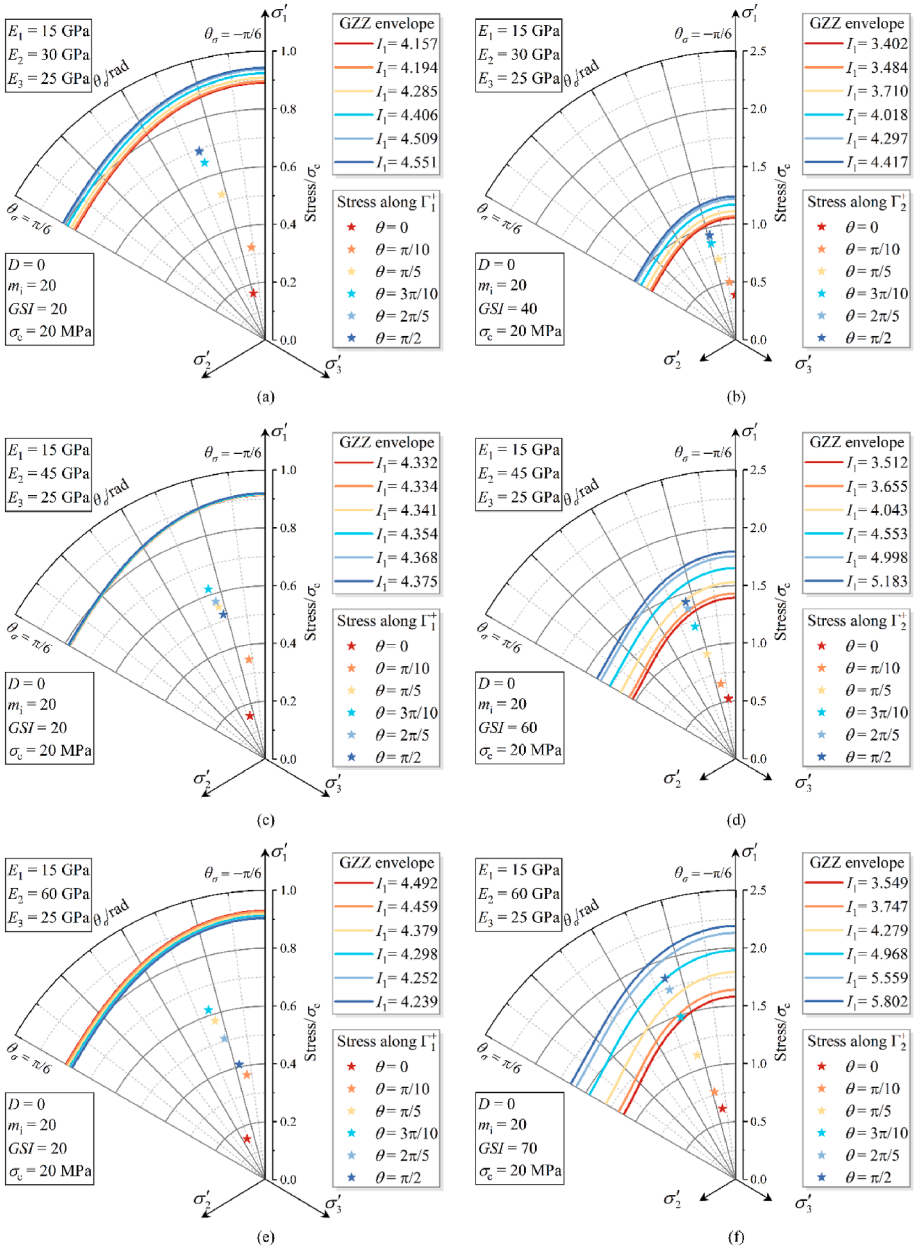


Fig. 13. Stress states in the π-plane for reinforced zone Ω_1 with different elastic moduli E_2: (a) $E_2 = 30$ GPa, boundary Γ_1^+, (b) $E_2 = 30$ GPa, boundary Γ_2^+, (c) $E_2 = 45$ GPa, boundary Γ_1^+, (d) $E_2 = 45$ GPa, boundary Γ_2^+, (e) $E_2 = 60$ GPa, boundary Γ_1^+, and (f) $E_2 = 60$ GPa, boundary Γ_2^+.

6.4. *Influence of the elastic modulus of surrounding rock in the reinforced zone Ω_2*

This section further investigates the impact of the elastic modulus of the surrounding rock in reinforced zone Ω_2 on the stress distribution and plastic zone extent. During calculations, the *GSI* of the surrounding rock in reinforced zone Ω_2 is set to 60, while other parameters remain consistent with those in Section 6.2. The proposed method is used to calculate the distribution patterns of the plastic zone and the failure approaching indices of the surrounding rock in reinforced zone Ω_2 for different values of E_3 (see Fig. 14). Furthermore, the stress distribution in the representative rock elements at the boundaries of the reinforced zones and the natural rock zone is illustrated in the π-plane (see Fig. 15), based on 6 observation points selected by rotating counterclockwise from the horizontal direction to the vertical direction.

As shown in Fig. 14a, the horizontal extent of the plastic zone at the arch lumbar decreases while the plastic zone at both the crown and bottom of the tunnel expands as E_3 increases, which results in a more uneven distribution and an increase in the geometric flattening of the plastic zone. Although the reinforcement in zone Ω_2 enhances the elastic modulus, it also amplifies the non-uniform effects of tectonic stress on the shape of the plastic zone. Therefore, to maintain the stability of the surrounding rock, the rock masses in the reinforced zone must possess higher strength to prevent asymmetrical development of the plastic zone. It is

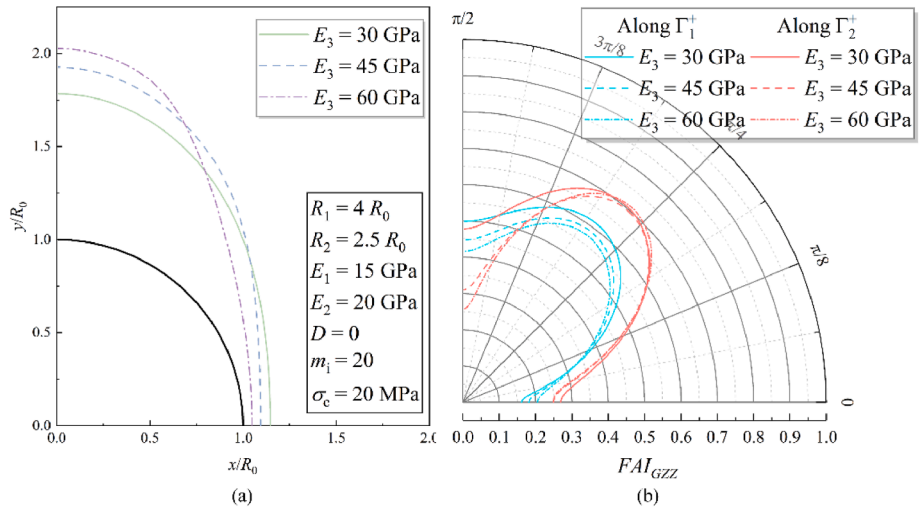


Fig. 14. Mechanical characteristics of surrounding rock in reinforced zone Ω_2 for different elastic moduli E_3: (a) shape and size of the plastic zones, and (b) failure approaching index FAI_{GZZ} at the reinforced zone boundaries.

demonstrated in Fig. 14b that increasing E_3 is beneficial for the stability of the surrounding rock in the natural rock zone and reinforced zone Ω_1. However, it fails to fully utilize the load-bearing capacity of the deeper rock mass, which may result in excessive loading on the shallow surrounding rock and the support structure.

It can be seen from Fig. 15 that increasing E_3 has a minimal impact on the stress level and distribution in the natural rock zone. The larger reinforced zone leads to a smaller disturbance range in the natural rock zone, which indicates that changes in the elastic modulus of the shallow surrounding rock have a limited effect on the stress distribution in the deeper surrounding rock. The increase in E_3 leads to a reduction in the stress level and concentration in the deeper surrounding rock. As a result, the shallow surrounding rock bears a greater load, which contradicts the tunnel support design concept that aims to fully utilize the deep surrounding rock to achieve a collaborative load-bearing capacity between both the deep and shallow surrounding rock. The larger elastic modulus of the shallow surrounding rock makes its stress state more complex. It can be found that the stress Lode angle θ_σ in the surrounding rock may exceed 0, which indicates a triaxial tensile stress condition. In such cases, tensile failure may occur if the strength of the surrounding rock undergoes unloading degradation, potentially leading to spalling or rock bursts.

7. Conclusions

In this paper, the functionally graded load-bearing concept based on grouting reinforcement of the surrounding rock is proposed. The mechanical model of the natural rock zone and FGRZs of deep-buried tunnels in high tectonic stress zones is constructed. The complex variable method is utilized to analyze the elasto-plastic mechanical response characteristics of the surrounding rock under gradient grouting. The effects of the strength and stiffness of the reinforced surrounding rock on the morphological characteristics and stability of the plastic zone under gradient grouting are investigated. The equivalent reinforcement method was adopted as an alternative to analyze the surrounding rock stability, while keeping the key parameters within a reasonable range. The effect of grouting on the improvement of the strength of the reinforced rock masses is approximated by increasing their *GSI* values. Although the proposed method can potentially address reinforced zones of arbitrary geometric complexity with further development, the current analysis is confined to circular reinforced zones due to space constraints. The conclusions drawn from this paper have the potential to guide the design of grouting reinforcement schemes in tunnel engineering. However, prior to practical application, further robust validation through model tests and on-site engineering monitoring data is still required. The main conclusions are as follows:

(1) In the surrounding rock of a deep-buried circular tunnel subjected to high tectonic stress, the plastic zones at the crown and bottom of the tunnel are widely distributed, exhibiting a strong linear correlation with the dimensionless factors K_σ and K_{SCF}. The surrounding rock at the arch lumbar of the tunnel is prone to a mixed tensile-compressive failure, and the circular grouting reinforcement zone plays a limited role in preventing instability and failure of the surrounding rock induced by unloading in extremely high tectonic stress tunnels.
(2) Grouting reinforcement has a significant impact on the mechanical properties and functional improvement of tunnels in weak, fragmented rock masses with low *GSI* values, which enhances the stability of the crown and bottom of the tunnels in high tectonic stress zones. The extent of the plastic zone in surrounding rock decreases non-linearly as the reinforcement strength increases, and it is not recommended to control the plastic zone through excessive grouting.
(3) An increase in the stiffness (elastic modulus) of the deep surrounding rock through grouting can, to some extent, transfer stress concentration from the shallow surrounding rock to the deeper surrounding rock, which helps to improve the uneven distribution of the plastic zone under high tectonic stress conditions. Furthermore, with constant strength

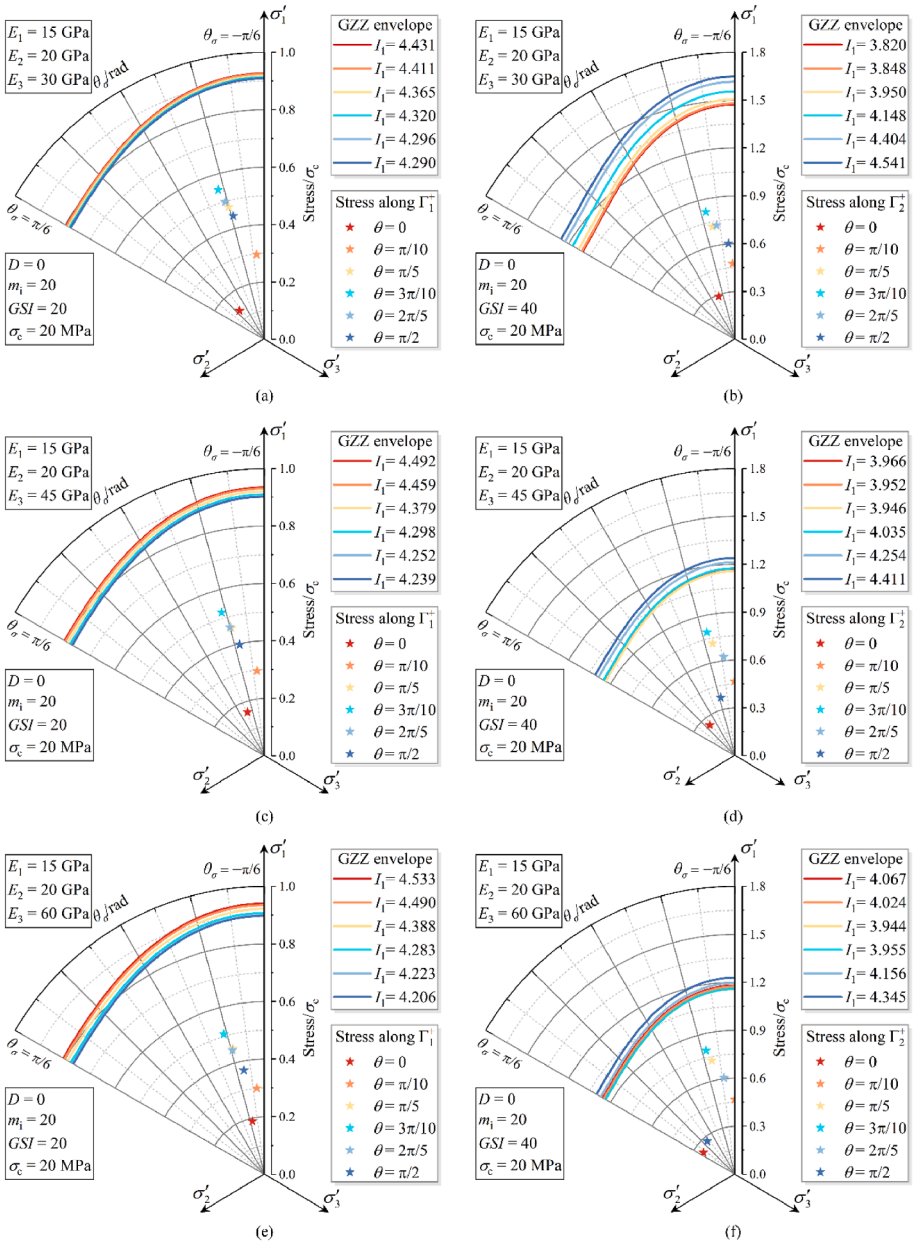


Fig. 15. Stress states in the π-plane for reinforced zone Ω_2 with different elastic moduli E_3: (a) $E_3 = 30$ GPa, boundary Γ_1^+, (b) $E_3 = 30$ GPa, boundary Γ_2^+, (c) $E_3 = 45$ GPa, boundary Γ_1^+, (d) $E_3 = 45$ GPa, boundary Γ_2^+, (e) $E_3 = 60$ GPa, boundary Γ_1^+, and (f) $E_3 = 60$ GPa, boundary Γ_2^+.

parameters in the reinforced zone, a reduction in the stiffness of the shallow surrounding rock can mobilize the collaborative load-bearing capacity of the deep surrounding rock, thereby reducing the extent of the plastic zone and enhancing the stability of the tunnel.

(4) Further exploration of grouting materials and techniques is necessary to establish FGRZs, where stiffness gradually increases and strength decreases from shallow to deep regions along the radial direction in the surrounding rock. This method utilizes the load-bearing capacity of the shallow rock and the pressure-regulation of the deep rock, actively controlling stress distribution and enhancing self-supporting characteristics. It is promising that functionally graded load-bearing effects will be achieved in both the natural rock and reinforced zones of deep-buried tunnels in high tectonic stress zones.

CRediT authorship contribution statement

Yaocai Ma: Software, Investigation, Formal analysis, Writing – original draft, Methodology, Funding acquisition, Data curation. **Hehua Zhu:** Supervision, Conceptualization, Project administration. **Wuqiang Cai:** Supervision, Funding acquisition, Writing – review & editing, Project administration, Conceptualization. **Chenlong Su:** Methodology, Data curation, Formal analysis. **Xiangyang Wei:** Investigation, Methodology, Formal analysis. **Zhi Zheng:** Investigation. **Jian Pu:** Investigation.

List of symbols

x, y	Cartesian coordinates in physical plane (z-plane)
r, α	Polar coordinates in physical plane (z-plane)
ρ, θ	Polar coordinates in image plane (ζ-plane)
u, v	Displacement along x-direction and y-direction
σ_c, m_i	UCS and material constant of intact rock
σ_1, σ_2, σ_3	Maximum principal stress, intermediate principal stress, and minimum principal stress
σ_x, σ_y, σ_z	Normal stress component along the x-direction, y-direction, and z-direction (tunnel axis direction) in the Cartesian coordinate system
σ_r, σ_α	Radial stress and tangential stress in the polar coordinate system
τ_{xy}, $\tau_{r\alpha}$	Shear stresses corresponding to the Cartesian and polar coordinate systems
σ_ρ, σ_θ, $\tau_{\rho\theta}$	Normal stress, tangential stress, and shear stress in the orthogonal curvilinear coordinate system
I_1, J_2, θ_σ	First invariant of stress, second invariant of the deviatoric stress, and stress Lode angle
m_b, s, a	Rock mass strength parameters related to *GSI* and *D*
GSI, *D*	Geological Strength Index and blast damage index
σ_x^∞, σ_y^∞	Horizontal (tectonic) geostress and vertical (gravitational) geostress
X_n, Y_n	Surface force components along the x-direction and y-direction
l, m	Direction cosines at the discussed boundary points
R_0, R_1, R_2	Radius of tunnel, reinforced zone Ω_1, and reinforced zone Ω_2, respectively
G_1, E_1, μ_1	Shear modulus, elastic modulus, and Poisson's ratio in natural rock zone
G_2, E_2, μ_2	Shear modulus, elastic modulus, and Poisson's ratio in reinforced zone Ω_1
G_3, E_3, μ_3	Shear modulus, elastic modulus, and Poisson's ratio in reinforced zone Ω_2
i	imaginary unit
z_1, z_2, z_3	Complex variables corresponding to points in natural rock zone, reinforced zone Ω_1, and reinforced zone Ω_2 in physical plane (z_1-, z_2-, and z_3-plane)
ζ_1, ζ_2, ζ_3	Complex variables corresponding to points in natural rock zone, reinforced zone Ω_1, and reinforced zone Ω_2 in image plane (ζ_1-, ζ_2-, and ζ_3-plane)
t_1, t_2, t_3	Boundary value of ζ_1, ζ_2, and ζ_3, respectively
r_1, r_2, r_3	Radius of the circle in ζ_1-, ζ_2-, and ζ_3-plane, respectively
$\omega_1(\zeta_1)$	Conformal mapping of natural rock zone in z_1-plane to the exterior region of a unit circle in ζ_1-plane
$\omega_2(\zeta_2)$	Conformal mapping of reinforced zone Ω_1 in z_2-plane to the annular region in ζ_2-plane
$\omega_3(\zeta_3)$	Conformal mapping of the elastic part of reinforced zone Ω_2 in z_3-plane to the annular region in ζ_3-plane
R, C_k	Conformal mapping coefficients of $\omega_1(\zeta_1)$
D_k	Conformal mapping coefficients of $\omega_2(\zeta_2)$
E_k	Conformal mapping coefficients of $\omega_3(\zeta_3)$
$\varphi_1(\zeta_1)$, $\psi_1(\zeta_1)$	Analytic functions representing the elastic stresses in natural rock zone
$\varphi_{11}(\zeta_1)$, $\varphi_{12}(\zeta_1)$	The two analytic functions that form $\varphi_1(\zeta_1)$ by linear superposition
$\psi_{11}(\zeta_1)$, $\psi_{12}(\zeta_1)$	The two analytic functions that form $\psi_1(\zeta_1)$ by linear superposition
$\varphi_2(\zeta_2)$, $\psi_2(\zeta_2)$	Analytic functions representing the elastic stresses in reinforced zone Ω_1
$\varphi_3(\zeta_3)$, $\psi_3(\zeta_3)$	Analytic functions representing the elastic stresses in the elastic part of reinforced zone Ω_2
a_k, b_k	Analytic function coefficients in $\varphi_{11}(\zeta_1)$ and $\psi_{11}(\zeta_1)$, respectively
c_0, c_k	Analytic function coefficients in $\varphi_{12}(\zeta_1)$
d_0, d_k	Analytic function coefficients in $\psi_{12}(\zeta_1)$
e_{-k}, e_0, e_k	Analytic function coefficients in $\varphi_2(\zeta_2)$
f_{-k}, f_0, f_k	Analytic function coefficients in $\psi_2(\zeta_2)$
g_{-k}, g_0, g_k	Analytic function coefficients in $\varphi_3(\zeta_3)$
h_{-k}, h_0, h_k	Analytic function coefficients in $\psi_3(\zeta_3)$
FAI_{GZZ}	Failure approaching index based on GZZ theory
K_σ, K_{SCF}	Dimensionless surrounding rock evaluation indices

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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References

Behnam, B., Fazlollah, S., Hamid, M., 2014. Prediction of plastic zone size around circular tunnels in non-hydrostatic stress field. J. Min. Sci. Technol. 24, 81–85.
Benato, A., Gaida, E., Oggeri, C., Oreste, P., 2023. Analysis of the importance of the filling material characteristics injected around the precast concrete lining in the microtunnelling technology. Geotech. Geol. Eng. 41, 2775–2785.
Bobet, A., Einstein, H.H., 2011. Tunnel reinforcement with rockbolts. Tunn. Undergr. Space Technol. 26, 100–123.
Cai, H., Lu, A.Z., Ma, Y.C., 2020. An analytical solution for shallow buried tunnel reinforced by point anchored rockbolts. Tunn. Undergr. Space Technol. 100, 103402.
Cai, W.Q., Zhu, H.H., Liang, W.H., Zhang, L.Y., Wu, W., 2021. A new version of the generalized Zhang–Zhu strength criterion and a discussion on its smoothness and convexity. Rock Mech. Rock Eng. 54 (8), 4265–4281.
Cai, W.Q., Zhu, H.H., Liang, W.H., 2022a. Three-dimensional stress rotation and control mechanism of deep tunneling incorporating generalized Zhang–Zhu strength-based forward analysis. Eng. Geol. 308, 106806.
Cai, W.Q., Zhu, H.H., Liang, W.H., Vu, B.T., Su, C.L., Zhang, K.S., Wang, X.J., 2022b. Three-dimensional forward analysis and real-time design of deep tunneling based on digital in-situ testing. Int. J. Mech. Sci. 226, 107385.
Cai, W.Q., Zhu, H.H., Liang, W.H., 2022c. Three-dimensional tunnel face extrusion and reinforcement effects of underground excavations in deep rock masses. Int. J. Rock Mech. Min. Sci. 150, 104999.
Cai, W.Q., Su, C.L., Zhu, H.H., Liang, W.H., Ma, Y.C., Xu, J.F., Xu, C., 2023. Elastic–-plastic response of a deep tunnel excavated in 3D Hoek–Brown rock mass considering different approaches for obtaining the out-of-plane stress. Int. J. Rock Mech. Min. Sci. 169, 105425.
Cai, Y., Jiang, Y.J., Djamaluddin, I., Iura, T., Esaki, T., 2015. An analytical model

considering interaction behavior of grouted rock bolts for convergence–confinement method in tunneling design. Int. J. Rock Mech. Min. Sci. 76, 112–126.
Chen, H.H., Zhu, H.H., Zhang, L.Y., 2022a. An analytical approach to the ultimate bearing capacity of smooth and rough strip foundations on rock mass considering three-dimensional (3D) strength. Compu. Geotech. 149, 104865.
Chen, H.H., Zhu, H.H., Zhang, L.Y., 2022b. A three-dimensional (3D) semi-analytical solution for the ultimate end-bearing capacity of rock-socketed shafts. Rock Mech. Rock Eng. 55, 611–627.
Chen, M., Lu, W.B., Zhang, W.J., Yan, P., Zhou, C.B., 2015. An analysis of consolidation grouting effect of bedrock based on its acoustic velocity increase. Rock Mech. Rock Eng. 48 (3), 1259–1274.
Chu, Z.F., Wu, Z.J., Liu, B.G., Liu, Q.S., 2019. Coupled analytical solutions for deep-buried circular lined tunnels considering tunnel face advancement and soft rock rheology effects. Tunn. Undergr. Space Technol. 94, 103111.
Chu, Z.F., Wu, Z.J., Liu, Q.S., Wen, L., Xu, X.Y., Wu, K., Sun, Z.Y., 2024. Viscos-elastic-plastic solution for deep buried tunnels considering tunnel face effect and sequential installation of double linings. Compu. Geotech. 165, 105930.
Cui, P., Ge, Y.G., Li, S.J., Li, Z.H., Xu, X.W., Zhou, G.G.D., Chen, H.Y., Wang, H., Lei, Y., Zhou, L.B., Yi, S.J., Wu, C.H., Guo, J., Wang, Q., Lan, H.X., Ding, M.T., Ren, J.J., Zeng, L., Jiang, Y.J., Wang, Y., 2022. Scientific challenges in disaster risk reduction for the Sichuan–Tibet Railway. Eng. Geol. 309, 106837.
Deng, S.H., Wang, X.L., Yu, J., Zhang, Y.C., Liu, Z., Zhu, Y.S., 2018. Simulation of grouting process in rock masses under a dam foundation characterized by a 3D fracture network. Rock Mech. Rock Eng. 51, 1801–1822.
Du, K., Tao, M., Li, X.B., Zhou, J., 2016. Experimental study of slabbing and rockburst induced by true-triaxial unloading and local dynamic disturbance. Rock Mech. Rock Eng. 49, 3437–3453.
Fang, Q., Zhang, D.L., Zhou, P., Wong, L.N.Y., 2013. Ground reaction curves for deep circular tunnels considering the effect of ground reinforcement. Int. J. Rock Mech. Min. Sci. 60, 401–412.
Feng, X.T., Tian, M., Yang, C.X., He, B.G., 2023. A testing system to understand rock fracturing processes induced by different dynamic disturbances under true triaxial compression. J. Rock Mech. Geotech. Eng. 15, 102–118.
Ghorbani, A., Hasanzadehshooiili, H., 2019. A comprehensive solution for the calculation of ground reaction curve in the crown and sidewalls of circular tunnels in the elastic-plastic-EDZ rock mass considering strain softening. Tunn. Undergr. Space Technol. 84, 413–431.
Gong, F.Q., Wu, W.X., Li, T.B., Si, X.F., 2019. Experimental simulation and investigation of spalling failure of rectangular tunnel under different three-dimensional stress states. Int. J. Rock Mech. Min. Sci. 122, 104081.
Gothäll, R., Stille, H., 2009. Fracture dilation during grouting. Tunn. Undergr. Space Technol. 24 (2), 126–135.
Gothäll, R., Stille, H., 2010. Fracture–fracture interaction during grouting. Tunn. Undergr. Space Technol. 25 (3), 199–204.
Guan, K., Zhu, W.C., Wei, J., Liu, X.G., Niu, L.L., Wang, X.R., 2018. A finite strain numerical procedure for a circular tunnel in strain-softening rock mass with large deformation. Int. J. Rock Mech. Min. Sci. 112, 266–280.
Guan, K., Zhu, W.C., Liu, X.G., Wei, J., 2020. Finite strain analysis of squeezing response in an elastic-brittle-plastic weak rocks considering the influence of axial stress. Tunn. Undergr. Space Technol. 97, 103254.
He, L., An, X.M., Zhao, Z.Y., 2015. Fully grouted rock bolts: an analytical investigation. Rock Mech. Rock Eng. 48 (3), 1181–1196.
He, M.M., Zhang, Z.Q., Zhu, J.W., Li, N., Li, G.F., Chen, Y.S., 2021. Correlation between the rockburst proneness and friction characteristics of rock materials and a new method for rockburst proneness prediction: field demonstration. J. Petrol. Sci. Eng. 205, 108997.
He, M.M., Zhang, Z.Q., Zhu, J.W., Li, N., 2022. Correlation between the constant m_i of Hoek–Brown criterion and porosity of intact rock. Rock Mech. Rock Eng. 55 (2), 923–936.
Hoek, E., Brown, E.T., 2019. The Hoek–Brown failure criterion and GSI–2018 edition. J. Rock Mech. Geotech. Eng. 11 (3), 445–463.
Jaeger, J.C., Cook, N.G.W., Zimmerman, R.W., 2007. Fundamentals of Rock Mechanics, fourth ed. Blackwell, Malden, USA.
Kargar, A.R., 2019. An analytical solution for circular tunnels excavated in rock masses exhibiting viscous elastic-plastic behavior. Int. J. Rock Mech. Min. Sci. 124, 104128.
Kargar, A.R., Haghgouei, H., Babanouri, N., 2020. Time-dependent analysis of stress components around lined tunnels with circular configuration considering tunnel advancing rate effects. Int. J. Rock Mech. Min. Sci. 133, 104422.
Kargar, A.R., Rahmannejad, R., Hajabasi, M.A., 2014. A semi-analytical elastic solution for stress field of lined non-circular tunnels at great depth using complex variable method. Int. J. Solid Struct. 51 (6), 1475–1482.
Kargar, A.R., Rahmannejad, R., 2015. An analytical solution for the ground reaction curve of brittle rocks, including gravity. Arab. J. Geosci. 8, 1479–1486.
Kang, H., Zhang, X., Si, L., Wu, Y., Gao, F., 2010. In-situ stress measurements and stress distribution characteristics in underground coal mines in China. Eng. Geol. 116 (3–4), 333–345.
Kim, B.K., Lee, I.M., Kee, T.H., Jung, J.H., 2019. Groutability enhancement by oscillatory grout injection: verification by field tests. Geomech. Eng. 18 (1), 59–69.
Lei, Q.H., Latham, J.P., Xiang, J.S., Tsang, C.H., 2017. Role of natural fractures in damage evolution around tunnel excavation in fractured rocks. Eng. Geol. 231, 100–113.
Li, P.F., Wang, F., Fang, Q., 2018. Undrained analysis of ground reaction curves for deep tunnels in saturated ground considering the effect of ground reinforcement. Tunn. Undergr. Space Technol. 71, 579–590.
Li, W.S., Jiang, B.Y., Gu, S.T., Yang, X.X., Shaikh, F.U.A., 2022. Experimental study on the shear behavior of grout-infilled specimens and micromechanical properties of grout-rock interface. J. Cent. South Univ. 29 (5), 1686–1700.
Ling, K., Wang, Y., Liu, D.Q., Guo, Y.P., Zhou, Z., Zhang, L.L., He, M.C., 2023. Experimental study on rockburst and spalling failure in circular openings for deep underground engineering. Rock Mech. Rock Eng. 56 (4), 2607–2631.
Lu, A.Z., Wang, S.J., Zhang, X.L., Zhang, N., 2018. Solution of the elasto-plastic interface of circular tunnels in Hoek–Brown media subjected to non-hydrostatic stress. Int. J. Rock Mech. Min. Sci. 106, 124–132.
Lu, A.Z., Zhang, N., Kuang, L., 2014. Analytic solutions of stress and displacement for a non-circular tunnel at great depth including support delay. Int. J. Rock Mech. Min. Sci. 70, 69–81.
Lv, Z.T., Yuan, S.H., Lin, H., 2023. Analytical stress solution for cold region tunnels with unequal ground stress and support delay under different frost heave conditions of surrounding rock. Cold Reg. Sci. Technol. 206, 103742.
Lyu, Z.T., Xia, C.C., Liu, W.P., 2020. Analytical solution of frost heaving force and stress distribution in cold region tunnels under non-axisymmetric stress and transversely isotropic frost heave of surrounding rock. Cold Reg. Sci. Technol. 178, 103117.
Ma, B.L., Zhang, K., Xiao, F.Y., Yu, B.S., Xue, X.R., Jiang, T., Zhang, H.Q., Zhang, Z.R., 2023a. Real-time estimating method on rock strength via MWD of roofbolter and its application to in-situ grouting quality evaluation. Geomech. Geophys. Geo-energ. Geo-resour. 9, 2363–8419.
Ma, Y.C., Cai, W.Q., Zhu, H.H., Su, C.L., Wei, X.Y., 2024. Plastic zone characteristics and rapid stability evaluation indexes of circular tunnels subjected to non-hydrostatic stress: a novel complex variable solution incorporating 3D rock mass strength. Tunn. Undergr. Space Technol. 151, 105868.
Ma, Y.C., Lu, A.Z., Cai, H., Zeng, X.T., 2022. Numerical applications of conformal mapping with explicit formulae based on the power series method. Int. Conf. Appl. Math. Model. Comput. Simulat.[C], Amsterdam: Advan. Transdisciplin. Eng. 20, 237–252.
Ma, Z.Y., Zuo, J.P., Zhu, F., Liu, H.Y., Xu, C.Y., 2023b. Non-orthogonal failure behavior of roadway surrounding rock subjected to deep complicated stress. Rock Mech. Rock Eng. 56 (9), 6261–6283.
Meng, F.Z., Wong, L.N.Y., Zhou, H., 2021. Rock brittleness indices and their applications to different fields of rock engineering: a review. J. Rock Mech. Geotech. Eng. 13, 221–247.
Osgoui, R.R., Oreste, P., 2010. Elasto-plastic analytical model for the design of grouted bolts in a Hoek-Brown medium. Int. J. Numer. Anal. Methods Geo-Mech. 34 (16), 1651–1686.
Setiawan, N.B., Zimmerman, R.W., 2020. A unified methodology for computing the stresses around an arbitrarily-shaped hole in isotropic or anisotropic materials. Int. J. Solid Struct. 199, 131–143.
Si, X.F., Gong, F.Q., 2020. Strength-weakening effect and shear-tension failure mode transformation mechanism of rockburst for fine-grained granite under triaxial unloading compression. Int. J. Rock Mech. Min. Sci. 131, 104347.
Su, C.L., Zhu, H.H., Cai, W.Q., Wu, W., Zhang, Q., 2023. Elastic perfect plastic rock mass analysis and equivalent GSI determination considering strain-softening behavior and three-dimensional strength. Tunn. Undergr. Space Technol. 142, 105422.
Sui, W.H., Liu, J.Y., Hu, W., Qi, J.F., Zhan, K.Y., 2015. Experimental investigation on sealing efficiency of chemical grouting in rock fracture with flowing water. Tunn. Undergr. Space Technol. 50, 239–249.
Sun, Z.Y., Zhang, D.L., Fang, Q., Liu, D.P., Dui, G.S., 2021. Displacement process analysis of deep tunnels with grouted rockbolts considering bolt installation time and bolt length. Compu. Geotech. 140, 104437.
Sun, Z.Y., Zhang, D.L., Fang, Q., Huangfu, N.Q., Chu, Z.F., 2023. Convergence-confinement analysis for tunnels with combined bolt–cable system considering the effects of intermediate principal stress. Acta Geotech 18, 3323–3348.
Sun, Z.Y., Zhang, D.L., Wang, J.C., Li, T.B., Hou, Y.J., 2024. Mechanical analysis of rock tunnels reinforced by grouted rockbolts considering the blasting damage effect. Rock Mech. Rock Eng. 57 (9), 7343–7369.
Trzeciak, M., Sone, H., 2022. Critical review of the Mogi failure criterion based on true-triaxial laboratory data analysis and theoretical considerations. Int. J. Rock Mech. Min. Sci. 159, 105220.
Wang, J., Liu, P., He, M.C., Tian, H.Z., Gong, W.L., 2023. Mechanical behaviour of a deep soft rock large deformation roadway supported by NPR bolts: a case study. Rock Mech. Rock Eng. 56 (12), 8851–8867.
Wu, W.X., Gong, F.Q., Ren, L., He, L., 2023. Strain rockburst failure characteristics and mechanism of high stress circular hard rock tunnel triggered by dynamic impact load. Int. J. Rock Mech. Min. Sci. 171, 105575.
Xie, H.P., Gao, M.Z., Zhang, R., Peng, G.Y., Wang, W.Y., Li, A.Q., 2019. Study on the mechanical properties and mechanical response of coal mining at 1000 m or deeper. Rock Mech. Rock Eng. 52 (5), 1475–1490.
Xie, X.Y., Zhai, J.L., Zhou, B., 2021. Back-fill grouting quality evaluation of the shield tunnel using ground penetrating radar with bi-frequency back projection method. Autom. ConStruct. 121, 103435.
Xiong, F., Jiang, Q.H., Ye, Z.Y., Zhang, X.B., 2018. Nonlinear flow behavior through rough-walled rock fractures: the effect of contact area. Compu. Geotech. 102, 179–195.
Xu, C., Xia, C.C., 2021. A new large strain approach for predicting tunnel deformation in strain-softening rock mass based on the generalized Zhang-Zhu strength criterion. Int. J. Rock Mech. Min. Sci. 143, 104786.
Xu, S.Q., Yu, M.H., 2006. The effect of the intermediate principal stress on the

ground response of circular openings in rock mass. Rock Mech. Rock Eng. 39 (2), 169–181.
Xu, X.Y., Wu, Z.J., Liu, Q.S., 2022. An improved numerical manifold method for investigating the mechanism of tunnel supports to prevent large squeezing deformation hazards in deep tunnels. Compu. Geotech. 151, 104941.
Xu, X.Y., Wu, Z.J., Weng, L., Chu, Z.F., Liu, Q.S., Wang, Z.Y., 2024. Investigating the impacts of reinforcement range and grouting timing on grouting reinforcement effectiveness for tunnels in fault rupture zones using a numerical manifold method. Eng. Geol. 330, 107423.
Yao, N., Deng, X.M., Luo, B.Y., Oppong, F., Li, P.C., 2023. Strength and failure mode of expansive slurry-inclined layered rock mass composite based on mohr–coulomb criterion. Rock Mech. Rock Eng. 56 (5), 3679–3692.
Yu, L.Y., Wu, D.Y., Su, H.J., Ge, S.T., Ju, M.H., Li, Y.H., Guo, J.F., 2025. Influences of fault slip induced by dynamic loading on the mechanical responses of tunnels. Tunn. Undergr. Space Technol. 160, 106533.
Zhang, D.L., Xu, T., Fang, H.C., Fang, Q., Cao, L.Q., Wen, M., 2022. Analytical modeling of complex contact behavior between rock mass and lining structure. J. Rock Mech. Geotech. Eng. 14 (3), 813–824.
Zhang, T., Yu, L.Y., Wei, J.B., Pu, H., Zhang, Q.H., Hu, L.H., Mi, X.Z., 2024a. Stress evolution in rocks around tunnel under uniaxial loading: insights from PFC3D-GBM modelling and force chain analysis. Theor. Appl. Fract. Mech. 134, 104728.
Zhang, K.S., Wu, W., Liu, Y.S., et al., 2024b. OCFMD: an automatic optimal clustering method of discontinuity orientation based on Fisher mixed distribution. Rock Mech. Rock Eng. 57 (3), 1735–1763.
Zhang, K.S., Wu, W., Liu, Y.S., Xie, T., Zhou, J.B., Zhu, H.H., 2024c. OCM: an intelligent recognition method of rock discontinuity based on optimal color mapping of 3D Point cloud via deep learning. Rock Mech. Rock Eng. 57 (7), 4873–4905.
Zhang, K.S., Wu, W., Zhu, H.H., Zhang, L.Y., Li, X.J., Zhang, H., 2020. A modified method of discontinuity trace mapping using three-dimensional point clouds of rock mass surfaces. J. Rock Mech. Geotech. Eng. 12 (3), 571–586.
Zhao, J., Hu, L., Feng, X.T., Xiao, Y.X., Guo, Y., 2023a. Shear failure mechanisms of sandstone subjected to direct, true triaxial and confining shear test conditions. Rock Mech. Rock Eng. 56 (9), 6889–6903.
Zhao, G., Yang, S., 2015. Analytical solutions for rock stress around square tunnels using complex variable theory. Int. J. Rock Mech. Min. Sci. 80, 302–307.
Zhao, Z.H., Li, C.L., Meng, Z., Liu, H., 2023b. Theoretical analysis of grouting reinforcement of surrounding rock with strength drop in deep chamber. Acta Mech. 234 (10), 4801–4819.
Zheng, Z., Deng, B., Li, S.J., Zheng, H., 2024a. Disturbance mechanical behaviors and anisotropic fracturing mechanisms of rock under novel three-stage true triaxial static-dynamic coupling loading. Rock Mech. Rock Eng. 57 (4), 2445–2468.
Zheng, Z., Xu, H.Y., Zhang, K., Feng, G.L., Zhang, Q., Zhao, Y.F., 2024b. Intermittent disturbance mechanical behavior and fractional deterioration mechanical model of rock under complex true triaxial stress paths. Int. J. Min. Sci. Technol. 34 (1), 117–136.
Zhu, H.H., Cai, W.Q., Liang, W.H., 2023. GZZ strength-based three-dimensional analysis theory and stress-controlled design method in deep tunneling. Chin. J. Rock Mech. Eng. 42 (1), 1–27 (in Chinese).
Zhou, X.P., Li, J.L., 2011. Hoek–Brown criterion applied to circular tunnel using elastoplasticity and in situ axial stress. Theor. Appl. Fract. Mech. 56 (2), 95–103.
Zhou, Y.L., Zhang, N., Ma, Y.C., Lu, A.Z., 2023. Stress analysis for an arbitrary-shaped tunnel in orthotropic rock masses based on a simple mapping function. Tunn. Undergr. Space Technol. 140, 105302.



Dr. Yaocai Ma obtained his PhD degree in Hydropower Engineering in June 2022 from the School of Water Resources and Hydropower Engineering at North China Electric Power University, Beijing, China, under the supervision of Professor Ai-Zhong Lu. Since August 2022, Dr. Ma has been conducting postdoctoral research in the Department of Geotechnical Engineering, College of Civil Engineering, Tongji University, Shanghai, China, working under the guidance of Professor He-Hua Zhu. He has received support from the China National Postdoctoral Program for Innovative Talents, the Shanghai Postdoctoral Excellence Program, and the National Natural Science Foundation of China for Young Scientists. He focuses on fundamental studies in underground tunnel engineering, including analytical methods for surrounding rock mechanics, strength theory of layered rock masses, and stress-control design in deep-buried tunnels.